THE BUSINESS INTELLIGENCE SEASON 9 COMPETITION

BUSINESS INTELLIGENCE COMPETITION ENTRY

TEAM : GÀ MỜ CÓ SỐ

Trần Ngọc Như Quỳnh	UEL
Trần Nguyên Diễm	UEL
Nguyễn Ngọc Kim Nga	UEL

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OVERVIEW

In the context of a rapidly shifting consumer landscape and intensifying competition within the food and beverage sector, data has become a critical asset for shaping informed, strategic decisions. This report is designed to serve as a comprehensive market intelligence document for Highlands Coffee, offering a multi-layered approach to understanding customer behavior, brand dynamics and organizational data infrastructure. As a market research analyst, the objective is not only to process and present data but also to interpret it in a way that aligns with the brand's business objectives and long-term positioning.

The report begins with a rigorous assessment of data quality, recognizing that the value of any analytical output is fundamentally dependent on the integrity of the underlying dataset. Through detailed preprocessing, redundant variables are identified and removed, inconsistencies in labeling are corrected, logical rule violations are addressed and missing values are imputed using appropriate statistical techniques. This phase is essential to ensure analytical accuracy and to support coherent integration across different data tables and dimensions.

Following the cleaning phase, the report introduces a conceptual data model that captures key relationships among business entities in a structured and normalized format. This data architecture serves as the backbone for all subsequent analytical activities. Parallel to this, the design of a role-based data access control system is proposed, tailored to the specific responsibilities and information needs of various departments and managerial levels. The aim is to enhance operational transparency while ensuring data security and governance, particularly in a multi-tiered organizational structure.

The latter half of the report transitions into strategic business analysis. A competitive landscape assessment is conducted to evaluate Highlands Coffee's brand performance in relation to key industry players, using customer perception metrics and brand funnel analytics. This analysis not only identifies areas of strength and weakness but also highlights opportunities for market repositioning. Furthermore, a comprehensive customer churn analysis is developed, leveraging behavioral, demographic, and perceptual data to identify patterns of disengagement. The insights derived from this analysis are operationalized through dynamic dashboards and advanced segmentation models that allow for the prediction of at-risk customer groups.

Ultimately, the report seeks to go beyond descriptive statistics by offering actionable insights that are rooted in business context. The final output represents an intersection between data science and strategic thinking, with the intention of supporting Highlands Coffee in refining its customer strategy, optimizing internal data governance and maintaining competitive relevance in a highly dynamic market environment.

PART 01

DATA QUALITY ASSESSMENT AND PREPROCESSING

1. File 2017Segmentation3685case.csv

- Data consists of 4,944 rows and 6 columns
- Check for duplicates: No duplicate rows
- Check for missing values: No null values in any column

→ Conclusion: Data is clean, no further processing required

- Save with new name: Segmentation_Cleaned.csv

2. File Companion.csv

- Data consists of 20739 rows and 4 columns
- Check for duplicates: Number of duplicate rows is 799 rows

→ Delete duplicate rows. Data after deletion is 19940 rows

- Check for missing values: No null values in any column

→ Conclusion: Data is clean, no further processing required

- Save as Companion_Cleaned.csv

3. File SA#var.csv

The initial data consists of 11,761 rows and 20 columns. After checking:

- No duplicate rows
- No outliers detected in the Age column

However, some columns have missing values or need to be processed, specifically as follows:

- **City:** 0 missing values

→ No processing required

- **Group_size:** 15 missing values

→ Fill in the value 0, because if there is no information, the default is to go alone

- **Age:** 9 missing values

→ Delete these rows because of missing age information - an important variable for analysis

- **Age#group:** 9 missing values

→ Delete missing rows because the age group cannot be determined

- **Age#Group#2:** 9 missing values

→ Delete missing rows because the age group cannot be determined

- **TOM:** 0 missing values

→ Normalize the labels "Other 1", "Other 2", "Other 3", "Other Branded Cafe Chain" to "Other"

- **BUMO:** 0 values missing

→ Merge "Other 1", "Other 2", "Other 3", "Other Branded Cafe Chain" into "Other"

- **BUMO_Previous:** 5,665 missing values

→ Assign "Don't have any brands" → "NoneBefore"; merge "Other" as above columns

- **MostFavourite:** 0 missing values

→ Combine "Other 1", "Other 2", "Other 3", "Other Branded Cafe Chain" → "Other"

- **Gender:** 0 missing values

→ No processing required

- **MPI#Mean:** 3,717 missing values

→ Fill in mode by Age#Group#2 and Occupation#group

- **MPI:** 3,717 missing values

→ Fill in mode by Age#Group#2 and Occupation#group

- **MPI#detail:** 3,685 missing values

→ Fill in mode by Age#Group#2 and Occupation#group

- **MPI#2:** 3,717 missing values

→ Delete column, due to duplicate with MPI

- **MPI_Mean_Use:** 6,974 missing values

→ Delete column, due to duplicate with MPI#Mean

- **Col:** 5,132 missing values

→ Delete column, not used in analysis

- **Occupation:** 0 missing values

→ Do not handle missing, used to standardize occupation labels for the group column below

- **Occupation#group:** 5,132 missing values

→ Fill in values using mode by Age

- **Year:** 5,132 missing values

→ Assign "Missing" value to empty cells

- Save with new name: Visitors_Cleaned.csv

4. File Brand_Image.csv

There are several data issues in this data table. These include:

- **Duplicate data**. In this table, by using the duplicated() function in the Pandas library (Python), there are 13,977 duplicate rows, accounting for approximately 2.2% of the table. However, this table reflects customers' brand-awareness surveys over multiple years, so a customer may legitimately participate multiple times with the same awareness level. Therefore, we will keep these duplicate rows.
- **Two columns contain identical data**. After checking each row in the Awareness and BrandImage columns, apart from their nulls, all values are identical. Therefore, we will drop the BrandImage column and retain Awareness (since BrandImage has more nulls), preserving all information while maintaining independent columns.

- **Fixing inconsistent labeling.** In Awareness, there are entries like Other, Other1, Other2, Other3 and Other Branded Cafe Chain. Although they all mean “Other brand,” their various forms cause inconsistency. we will standardize them all to Other to ensure label consistency.
- **Missing data**. In Brand_Image, only one column has missing values: 397 rows (0.06% of the table). These are MCAR (Missing Completely At Random), since they don’t depend on any other variable. We will impute them using the column’s mode, preserving critical customer-awareness information.

5. File Competitor database_xlnm#_FilterDatabase.csv

There are no data issues such as duplicates, logical errors, inconsistent labeling, or missing values.

Save with new name: Competitor_Cleaned.csv

6. File NeedstateDayDaypart.csv

There are several data issues in this data table including:

- **Duplicate data.** In this table, there are 6,258 duplicate rows, accounting for 8.2% of all rows. This does not stem from system errors or data-entry issues but reflects real behavior - customers visiting the café in different cities or years - so we will keep these duplicates.
- **Missing data.** Two columns have missing values: Day#Daypart has 11,907 missing rows (15.8%) and NeedstateGroup also has 11,907 missing rows (15.8%). Both are MAR (Missing At Random), since their absence depends on the observed Needstates field. We will fill NeedstateGroup by mapping it from Needstates, preserving data accuracy; and fill Day#Daypart by grouping on NeedstateGroup and imputing each group’s mode.

Save with new name: Need_State_Day_Cleaned.csv

7. File Brandhealth.csv

- **Removing Duplicates:** no further duplicates remained in the dataset.
- **Logical Consistency Check:**
- Spending must imply awareness and trial: If a respondent reported non-zero spending, we assumed they must be aware of and have tried the brand. We identified nine violations where either awareness or trial was missing and corrected them by assigning the brand name to the relevant missing field.
- **Visits require a product trial:** A customer cannot have repeated visits without having first tried the product. Missing Trial entries in these cases were filled accordingly.
- **Purchase intent without engagement:** Although no inconsistencies were found, we verified that all responses showing purchase path attribution (PPA) were backed by either spending or visit history.

Save as Brand_Health_Cleaned.csv

- **Recent usage (P1M/P3M) without trial:** Customers who recently interacted with a brand were expected to have tried it. Missing trials in these records were corrected.
- **Daily/Weekly behavior missing monthly records:** If a respondent indicated they visited a brand daily or weekly, we expected corresponding entries in monthly usage (P1M, P3M). These were added where absent.
- **Trial without awareness:** Trial implies a level of brand recognition, so we assumed that awareness should exist if trial data was available. We aligned missing Awareness values with the Trial brand where needed.
- **Handling missing values:** After completing the logic checks, we observed that several key columns in the Brand_Health dataset still had substantial missing values. To ensure data consistency, we applied a set of rule-based imputations. Categorical variables like Awareness, Spontaneous, and usage indicators such as P3M, P1M, Trial, Weekly, and Daily were filled with "Unknown" or "No" where appropriate. The column Brand_Likability was renamed to Brand_Liking and missing values were set to "No". For numerical fields, we standardized Fre#visit and Spending, replacing nulls with 0, and created a derived field Spending_use. Additional logic ensured that if users reported recent usage (P3M ≠ No) but had no visit count, at least one visit was assumed. Brand names were also harmonized by grouping all "Other" labels under a single "Other" category. For perception metrics like NPS#P3M, NPS#P3M#Group and Comprehension, we used statistical imputation: filling missing values based on the mode within relevant groups or the dataset as a whole. This process allowed us to preserve analytical validity while ensuring the dataset was complete and logically coherent for further analysis.

8. File Dayofweek.csv

- **Removing Duplicates:** found and eliminated 37 duplicate rows, ensuring each observation represented a unique visit behavior.
- **Logical Consistency Check:** two major logic issues were addressed:
- Visit count present but weekday missing: For cases where visit frequency (Visit#Dayofweek) was recorded but the corresponding Dayofweek was missing, we inferred the most likely day using the statistical mode of similar entries. However, 50 rows lacked a clear dominant pattern and were flagged for future inspection.
- Weekday specified but visit count missing: If a day of the week was recorded, at least one visit should be assumed. We assigned a default value of 1 to these cases.

Save with new name: Day_of_week_Cleaned.csv

- **Handling missing values:** After fixing logic issues in the Day_of_week dataset, we addressed missing values in the Weekday#end column by categorizing days into weekdays and weekends based on the Dayofweek field. This step helped ensure consistent time-based grouping and improved data completeness.

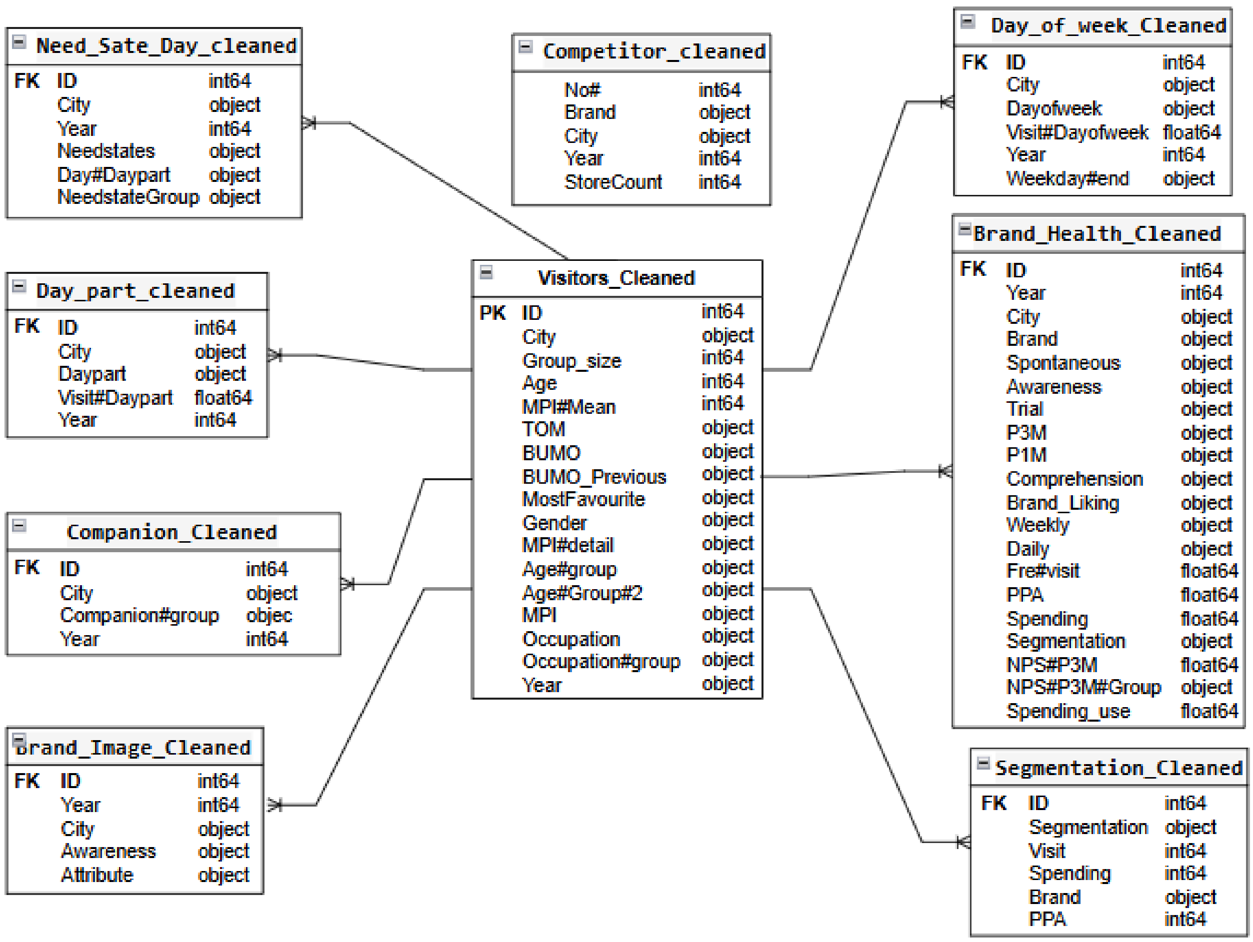
9. File Daypart.csv

- **Removing Duplicates:** did not contain duplicate entries, which allowed us to focus entirely on behavioral logic checks.
- **Logical Consistency Check and handling missing values:** Missing Daypart values were filled using the most frequent time slot corresponding to similar visit counts. For entries with a specified Daypart but no visit data, a default value of 1 was assigned, assuming at least one visit took place.

Save with new name: Day_part_Cleaned.csv

PART 02

DATA MODELING DIAGRAM



Entity Relationship Diagram (ERD)

Need_Sate_Day_Cleaned stores the various need-state groups that each visitor can select in the survey. The relationship to Visitors_Cleaned is **one-to-many** because a single visitor ID may be associated with multiple need-state records, reflecting that one person can simultaneously belong to several need-state categories.

Day_part_Cleaned captures the actual time slots when a visitor attended the survey. Although any single visit has just one time slot, a visitor may return at different hours on different occasions, so we model this as **one-to-many**: one visitor ID links to multiple Visit#Daypart entries, allowing us to analyze visit frequency and timing.

Day_of_week_Cleaned records the day of the week on which each survey response occurred. Since a visitor can participate in surveys on different days (for example, across several weeks or multiple times in one week), the association with Visitors_Cleaned is also **one-to-many**, enabling us to capture each visit under its specific weekday.

Companion_Cleaned describes the group of companions (alone, friends, family,...) for each survey instance. A visitor might come alone one time and with friends or family on another, so the link between Visitors_Cleaned and Companion_Cleaned remains **one-to-many**, with one visitor ID mapping to multiple companion-group records.

Brand_Image_Cleaned holds visitors' ratings of brand image attributes (Awareness, Attribute,...). Because a single visitor can evaluate many different facets of a brand's image—either in one survey or across several - the relationship is **one-to-many**, with each visitor ID giving rise to multiple image-attribute records.

Brand_Health_Cleaned stores brand health metrics such as Trial, P3M, and NPS. One visitor may supply a range of metrics for a brand, so we again use a **one-to-many** relationship to record each metric in its own row.

Segmentation_Cleaned captures segmentation, visit count, spending, brand choice, and PPA for each visitor. Since one person can fall into multiple segments or generate several measurements over time, the link to Visitors_Cleaned is **one-to-many**, allowing multiple segmentation records per visitor.

Competitor_Cleaned is an **independent table** that aggregates store counts by brand, region, and year. It has no foreign key to Visitors_Cleaned, because it reflects the overall branch network of each brand rather than individual customer responses.

All of these dimension tables conform to Third Normal Form (3NF): every column holds atomic values, all attributes depend fully and directly on the table's primary key, and there are no transitive dependencies. This design minimizes redundancy, maintains consistency, and simplifies future updates.

PART 03

ROLE-BASED DATA ACCESS CONTROLS

1. Role Definition and Permission Assignment

The table below describes the primary tasks, data needs, and corresponding data tables that each department needs to access to perform its job functions.

Department	Main mission	Data needs	Corresponding data table
Board of Directors (BOD)	Strategic direction, monitoring brand health and system-wide performance	- Macro indicators - System-wide performance - Brand KPIs - Overall benchmarking	**Full access to all data tables**
HR / People & Reward Analytics	Human resources management, performance tracking for rewards/penalties	- Employee information - Attendance records (working days, leaves) - Individual performance - Punctuality, complaints, error rates	**No data access**
Finance and Accounting	Financial control, cost measurement, profit forecasting	- Financial transactions - Costs and budgeting - Resource allocation and traffic	**No data access**
Marketing	Brand strategy, communications, market/customer analysis	- Customer attitude - Brand health metrics (Awareness, Usage, Preference, etc.) - Customer, market, and competitor information	**Full access to all data tables**

Department	Main mission	Data needs	Corresponding data table
CRM Lead	Customer relationship strategy and retention	- Customer behavioral data and satisfaction - Customer interaction history - Segmentation and loyalty metrics	**All tables except: Competitor, Companion, NeedstateDay, Daypart**
Sales Operation	Sales operations management, performance monitoring	- Sales revenue and costs - Individual and team KPIs	**Competitor, Brand_Image, Brand_Health tables**
Regional Managers	Regional supervision, planning, performance tracking	- Store performance per region - Customer behavior by region - Local competitive trends	**Competitor, Brand_Image, Brand_Health, Visitors, Day_of_week, Day_part tables after filtering by region**
Store Managers	Store-level operations, finance, staff and promotion management	- Real-time store-level performance (e.g., sales, inventory, staff) - Real-time customer behavior in-store	**No access granted (due to lack of real-time data)**

2. Data access matrix

Access Level Definitions

- **Full Access:** Granted full access to all records and fields within the data table.
- **No Access:** Access denied due to lack of relevance to the role's responsibilities.
- **Filtered Access:** Granted partial access to records filtered by relevant criteria such as region, store ID, or role-specific scope.

Department / Data table	Board of Directors (BOD)	HR/People & reward analytics	Finance and Accounting	Marketing
Segmentation	Full Access	No Access	No Access	Full Access
Brand_Image	Full Access	No Access	No Access	Full Access
Brand_Health	Full Access	No Access	No Access	Full Access
Companion	Full Access	No Access	No Access	Full Access
Competitor	Full Access	No Access	No Access	Full Access
Day_of_week	Full Access	No Access	No Access	Full Access
Day_part	Full Access	No Access	No Access	Full Access
Need_state_Day	Full Access	No Access	No Access	Full Access
Visitors	Full Access	No Access	No Access	Full Access

2. Data access matrix

Data table \ Department	CRMLead	Sales Operation	Regional Managers	Store Managers
Segmentation	Full Access	No Access	No Access	No Access
Brand_Image	Full Access	Full Access	Filtered Access	No Access
Brand_Health	Full Access	Full Access	Filtered Access	No Access
Companion	No Access	No Access	No Access	No Access
Competitor	No Access	Full Access	Filtered Access	No Access
Day_of_week	Full Access	No Access	Filtered Access	No Access
Day_part	No Access	No Access	Filtered Access	No Access
Need_state_ Day	No Access	No Access	No Access	No Access
Visitors	Full Access	No Access	Filtered Access	No Access

3. Reasons for each role

Role	Justification for Data Access Level
Board of Directors (BOD)	With responsibility for strategic decision-making and overall oversight, the BOD requires full access to assess performance and risks at the system-wide level.
HR / People & Reward Analytics	The available data is not aligned with HR's functional responsibilities. Granting access could pose risks of disclosing information beyond their scope of personnel management.
Finance & Accounting	Access to brand or customer behavior data is not necessary for financial duties. Restricting access helps prevent unnecessary exposure to sensitive information outside financial scope.
Marketing	Broad access is required to analyze brand performance, customer behavior, and market dynamics, supporting the role's strategic planning and campaign optimization functions.
CRM Lead	Access is granted only to data supporting customer retention efforts. Restricting access to unrelated tables minimizes the risk of leaking strategic information.
Sales Operation	Limited access to performance and competitive data ensures alignment with sales operations duties, while avoiding exposure to unrelated datasets.
Regional Managers	Access is filtered by region to enable effective supervision and planning, while avoiding unnecessary access to data outside their geographic scope.
Store Managers	Access is not granted due to the absence of real-time data support. This restriction prevents unnecessary data exposure beyond the store-level scope.

PART 04

COMPETITIVE LANDSCAPE ANALYSIS AND BRAND POSITIONING FOR HIGHLAND COFFEE

Competitive Landscape Analysis

1. Analysis by brand recognition rate

Count by Awareness

Count by Sponaneous

Key message:

- Highlands ranks 2nd in total brand awareness.
- But leads in spontaneous brand recall.

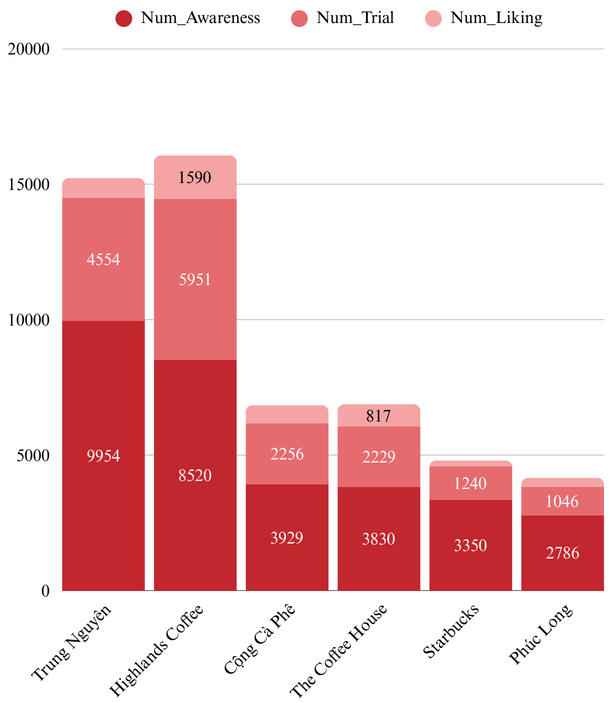
Strategic significance:

- Highlands has the highest “salience” in the minds of customers.
- This is a strong brand advantage, difficult to build and easy to maintain market share.

→ ***To provide a more focused analysis, the team has identified five key competitors of Highlands based on market context and data: Trung Nguyên, Cộng Cà Phê, The Coffee House, Starbucks and Phúc Long.***
In the following visualizations, only these main competitors will be filtered for comparison.

Competitive Landscape Analysis

1. Analysis by brand recognition rate

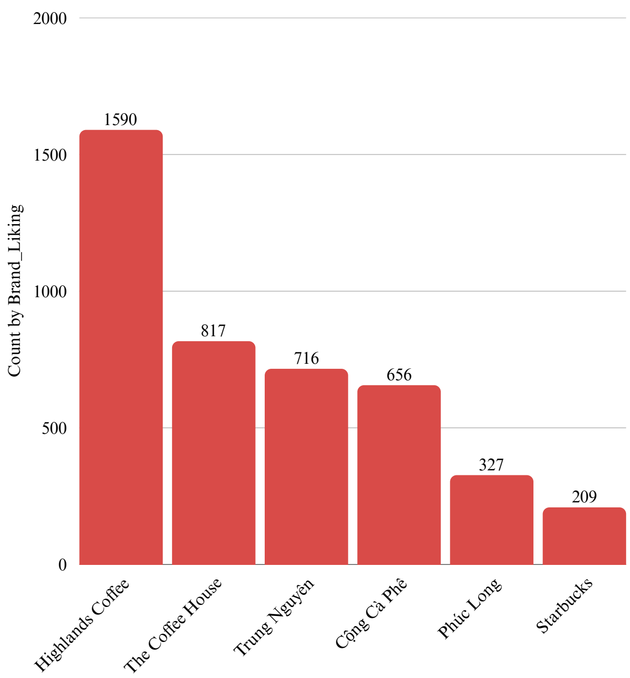


Key message:

- Highlands has the best conversion performance: from know → try → like.
- Trung Nguyen is well known, but the like rate is lower than Highlands.

Strategic significance:

- Highlands builds its brand with experience, not just promotion.
- Competitors such as Cong Cafe, The Coffee House only achieve average Awareness and Trial.



Key message:

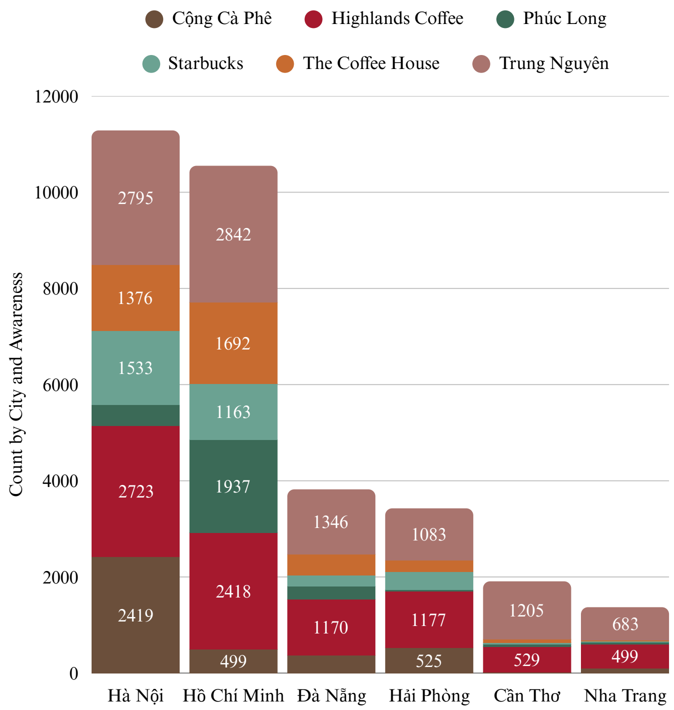
- Highlands overwhelms its competitors in terms of number of fans.
- Double that of The Coffee House, nearly 8 times that of Starbucks.

Strategic significance:

- Highlands is the brand that best connects with users' emotions.
- This is the factor that retains loyal customers, increasing average spending.

Competitive Landscape Analysis

1. Analysis by brand recognition rate

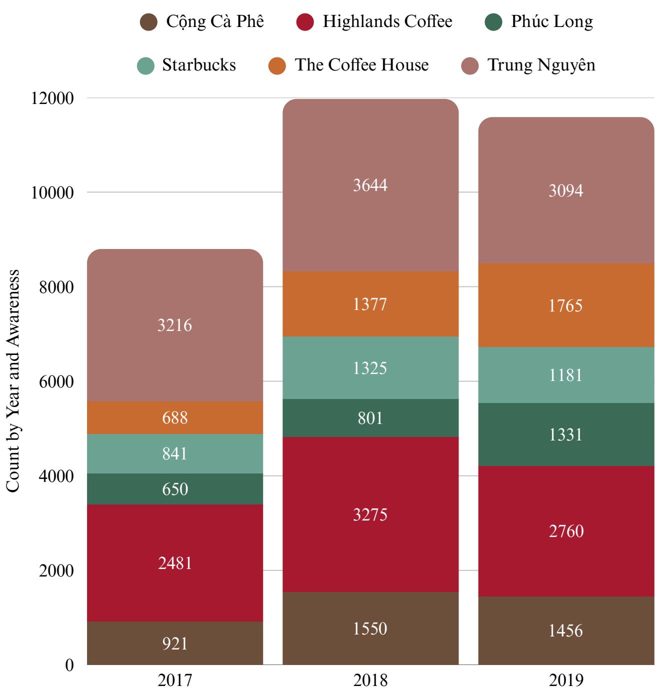


Key message:

- Highlands is present in all major cities, ranking top in HCM and Hanoi.
- Competitors such as Cong Cafe and The Coffee House focus more on Hanoi.

Strategic significance:

- Highlands is dominating both the Northern and Southern markets, demonstrating its ability to expand.
- A national competitor, not a regional brand.



Key message:

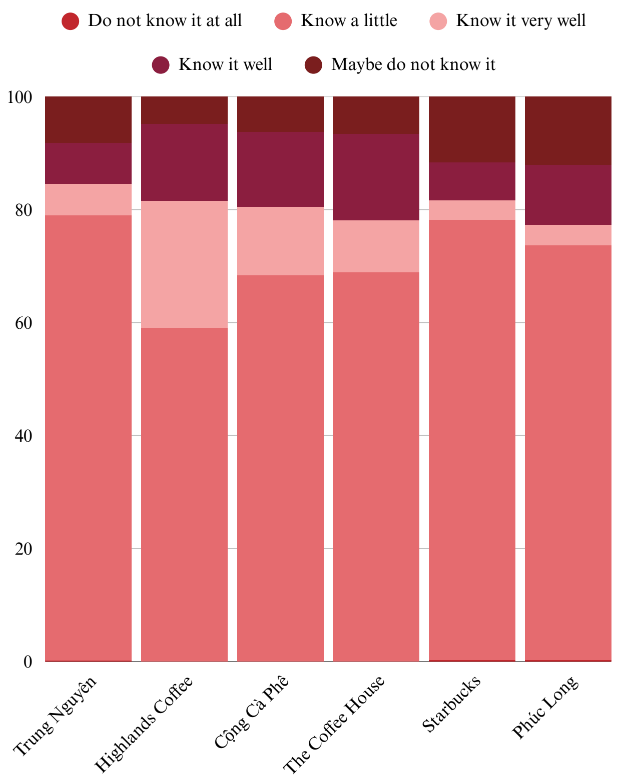
- Highlands has grown consistently over the years (2017–2019), surpassing Trung Nguyen since 2018.
- Other brands have remained stable or stagnant.

Strategic significance:

- Highlands is investing in effective and targeted marketing.
- Other brands need to improve their recognition growth rate.

Competitive Landscape Analysis

2. Analysis based on conversion stage



Key message:

- Highlands Coffee is the brand with the highest rate of consumers "know it very well" among the brands.
- Trung Nguyen and Cong Ca Phe have a fairly high level of recognition, but the "know it very well" rate is lower than Highlands.
- Brands such as Phuc Long and Starbucks have a fairly high rate of "don't know or only know a little", showing that brand recognition is not strong enough in the large market.

Strategic significance:

- Highlands is not only known by many people, but is also a well-remembered and well-understood brand, reflecting the effectiveness of communication and brand positioning.
- Starbucks, although a global brand, is still a rather obscure name in the minds of mass customers in the Vietnamese context.
- Trung Nguyen still has good recognition, but is not as prominent as Highlands in the "know it very well" part.

Three stages in the brand awareness journey of Highlands' five main competitors

- **Cộng Cà Phê**

3929 Num_Awarenees	**2256 Num_Trial**	**656 Num_Liking**

- **Highlands Coffee**

8520 Num_Awarenees	**5951 Num_Trial**	**1590 Num_Liking**

Competitive Landscape Analysis

2. Analysis based on conversion stage

- **Phúc Long**

2786 Num_Awarenees	1046 Num_Trial	327 Num_Liking

- **Starbucks**

3350 Num_Awarenees	1240 Num_Trial	209 Num_Liking

- **The Coffee House**

3830 Num_Awarenees	2229 Num_Trial	817 Num_Liking

- **Trung Nguyên**

9954 Num_Awarenees	4554 Num_Trial	716 Num_Liking

→ Highlands leads in brand awareness and shows strong conversion across all stages.

→ Cộng Cà Phê shows an impressive conversion from awareness to trial.

→ The Coffee House excels in generating liking post-trial.

→ Starbucks and Phúc Long have low conversion rates

→ Trung Nguyên has strong awareness but low customer liking rate.

Conclusions

1. Brand Awareness:

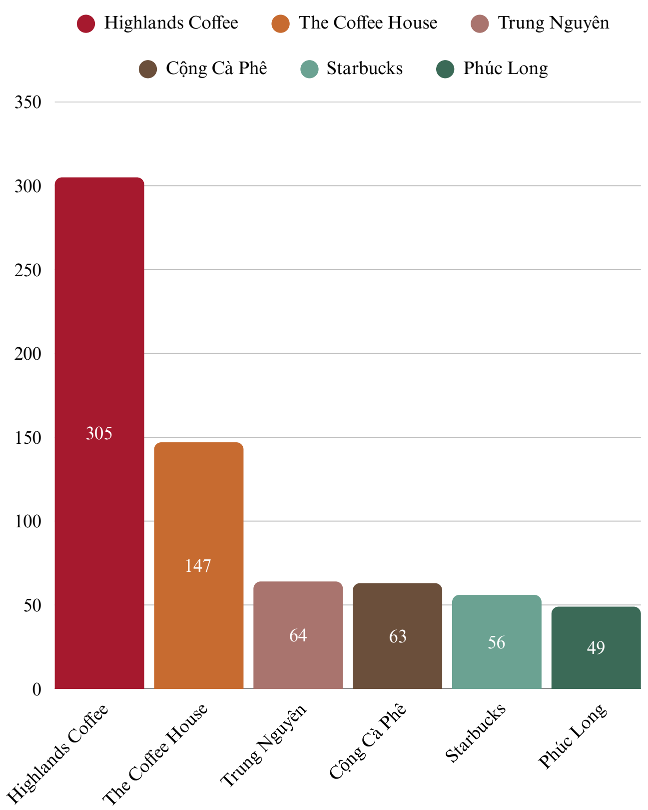
Highlands Coffee ranks second in awareness but leads in spontaneous recall and "know very well".

→ Positioning: A popular, easily recalled brand with strong mental presence.

Competitive Landscape Analysis

2. Conversion Efficiency:
Highlands excels at converting awareness → trial → liking, leading all competitors in brand liking.
→ Positioning: Not just well-known, but deeply loved.
3. Brand Liking:
Highlands is the most liked coffee brand – double The Coffee House and 8× Starbucks.
→ Positioning: A brand with strong emotional connection among Vietnamese consumers.
4. Market Presence & Growth:
Highlands has strong presence in both major cities and provinces, and grows steadily over time, surpassing Trung Nguyên since 2018.
→ Positioning: A nationwide, dynamic, and expanding brand.
5. Competitor Comparison:
Trung Nguyên: High awareness, lower trial and liking.
The Coffee House, Cộng: Medium awareness, urban-focused, moderate liking.
Starbucks, Phúc Long: Low liking and presence, struggling to build loyalty.

3. Analysis of market presence



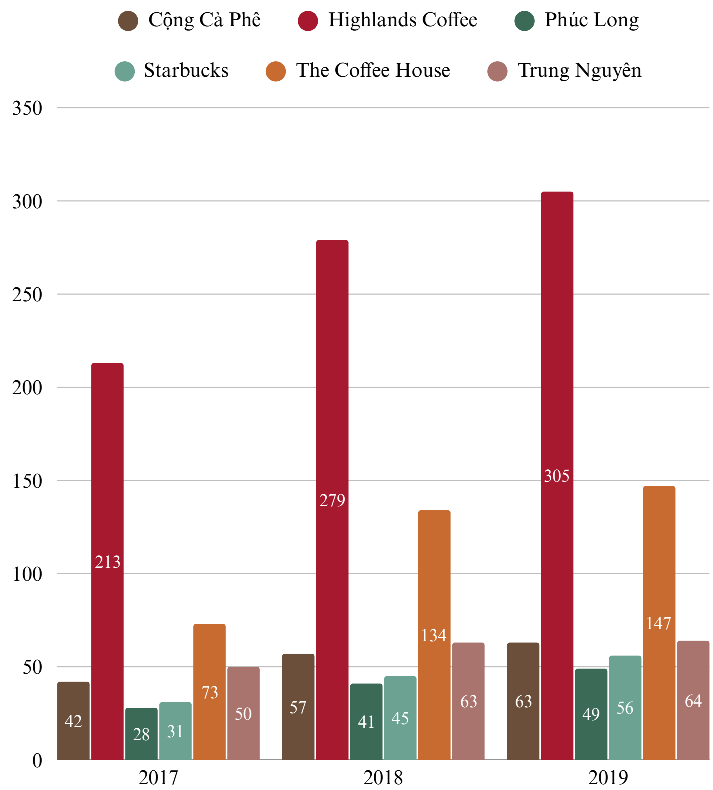
Number of stores by brand

→ Highlands Coffee dominates in store count, doubling The Coffee House.

→ Other brands have significantly fewer stores.

Competitive Landscape Analysis

3. Analysis of market presence

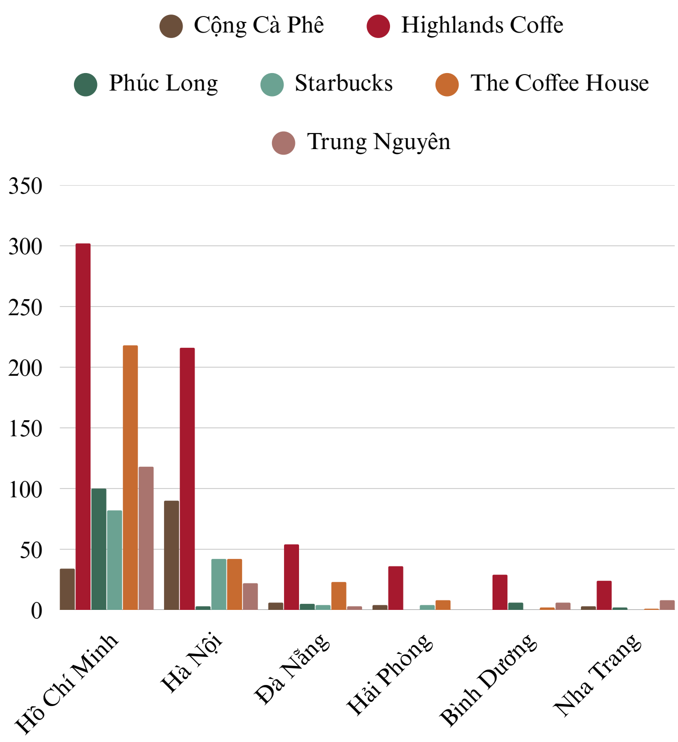


Number of stores by brand by year

→ Highlands Coffee shows consistent and strong expansion from 2017 to 2019.

→ The Coffee House also grows rapidly but still trails Highlands.

→ Other brands show slight or stagnant growth.



Number of stores by brand by city

→ Highlands dominates both Ho Chi Minh City and Hanoi.

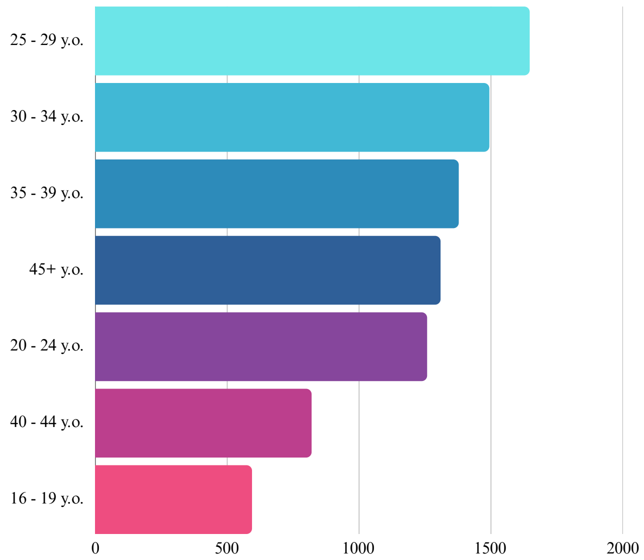
→ The Coffee House and Trung Nguyên are mostly present in the two major cities.

→ Highlands is one of the few brands with coverage in smaller cities like Hai Phong, Binh Duong, and Nha Trang.

→ Highlands Coffee not only leads in store count but also expands faster and has broader geographic coverage than competitors — a significant advantage in market presence and customer reach.

Highlands Coffee brand positioning

1. Age Distribution



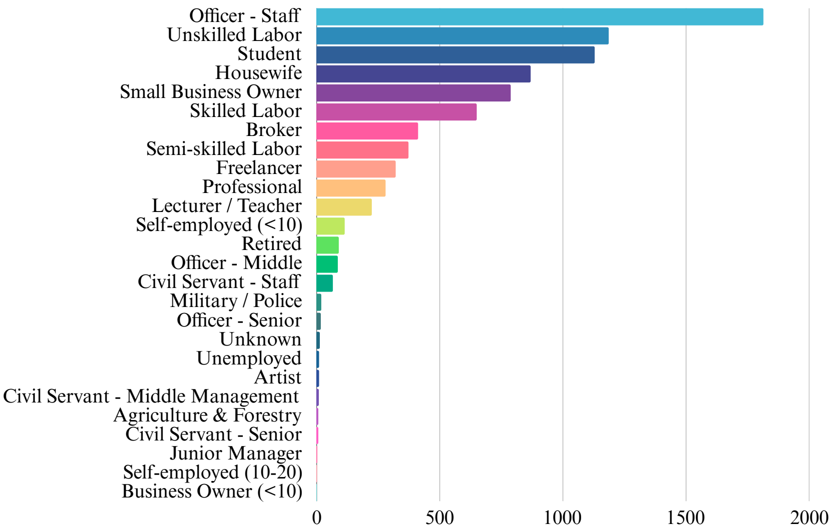
→ The largest age group is 25–29 years old

→ Customers aged 30–39 also make up a large share, indicating a stable working-age customer base.

→ The 45+ group is also significant, proving Highlands appeals beyond youth.

→ The 16–24 group is smaller but holds potential for future targeting.

2. Occupation Distribution



→ Office staff dominate, forming the core customer base.

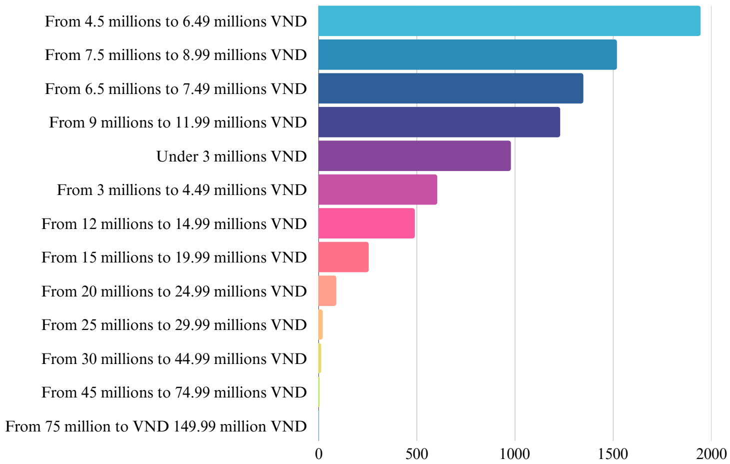
→ Unskilled labor and students are the next largest segments.

→ Housewives and small business owners are notable segments.

→ Professionals, managers, and senior civil servants are underrepresented.

Highlands Coffee brand positioning

3. Income Distribution



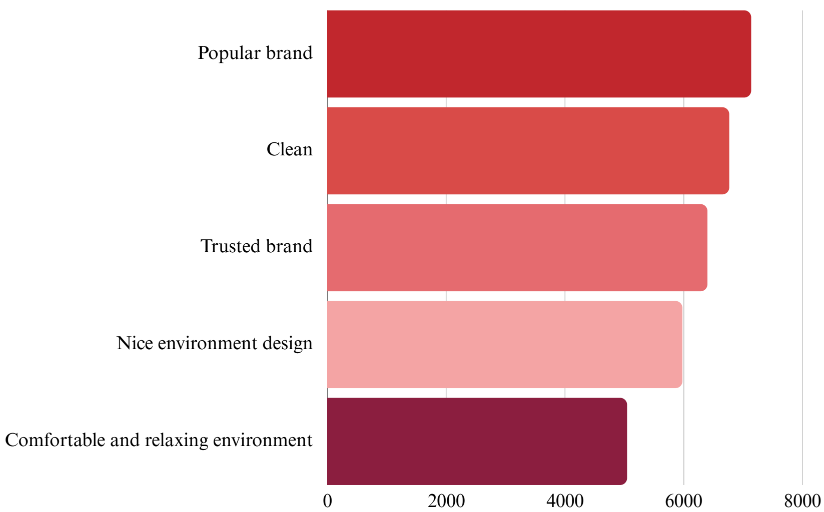
→ Most customers earn from 4.5 – 9 million VND/month, placing Highlands in the mid-range market.

→ The 9 – 12 million group also represents a considerable share.

→ Nearly 1,000 customers earn under 3 million, showing Highlands' accessibility to low-income groups.

→ High-income customers (over 15 million VND) are a small minority.

4. Reasons to Choose Highlands



Top 5 user-rated attributes of Highlands Coffee

→ "Popular brand" ranks first, confirming Highlands' strong brand presence and recognition.

→ Nice environment and relaxing space are valued but less emphasized, suggesting room for spatial improvement.

Final Summary

- Customers are mainly aged 25–39, representing a stable, working-age coffee-drinking demographic.
- Office workers, unskilled laborers, and students are the dominant occupation groups.
- Most customers earn 4.5–9 million VND/month, reflecting a clear mid-tier market positioning.
- Brand strength, trustworthiness, and cleanliness are top reasons. Ambience-related attributes show room for emotional engagement improvement.

Strategic Suggestions

- Enhance in-store environment to strengthen emotional connection with customers, especially office workers.
- Maintain strong brand image with a focus on trustworthiness and cleanliness – the top three attributes that attract most customers.
- Leverage loyalty programs or personalized promotions to retain the core 25–39 age group.

PART 05

CUSTOMER CHURN ANALYSIS AND DASHBOARD FOR HIGHLANDS COFFEE

Introduction and methodology

Customer churn, defined as the phenomenon of brand abandonment, is a critical indicator of customer engagement and long-term brand health. Monitoring churn enables companies to detect emerging risks, fine-tune strategic initiatives and optimise customer-care resources.

In this study, a customer is classified as churned when activity is recorded within the past three months (P3M not equal to "No") yet completely absent in the most recent month (P1M equal to "No"). This rule is encoded in the is_churned field, which places customers into three categories: churned, active and insufficient-data.

To quantify churn, two custom measures are created. The first, Churned Cust, counts customers in the churned category. The second, Not Churned Cust, counts customers who remain active. These measures form the quantitative backbone of subsequent analyses.

The dataset is then explored across several dimensions, including demographics, brand perception variables, consumption behavior, and temporal usage patterns. A drill-down approach is employed to uncover underlying drivers at each hierarchical level, thereby highlighting priority customer segments for targeted retention strategies.

Overview churn customer of Highland Coffee

2694	1452
Not Churned Customer	Churned Customer

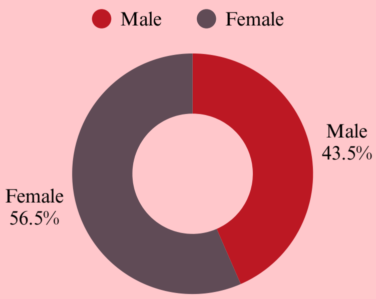
The total number of customers currently classified as active is 2,694, while 1,452 individuals have been identified as having disengaged from the brand. This corresponds to an estimated churn rate of approximately **35%**, a figure that merits particular attention in the context of brand performance.
Such a level of attrition is considerable, particularly for a brand with strong market visibility and widespread consumer reach like Highlands Coffee. Rather than representing an isolated fluctuation, this churn rate may point to more systemic challenges in maintaining customer engagement over time. It could also suggest a gradual erosion of brand loyalty, inconsistencies in service delivery or a potential disconnect between customer expectations and the actual experiences provided.
This emerging pattern highlights the importance of adopting a more nuanced and comprehensive view of churn dynamics, which will be further examined in subsequent sections of the report.

Multi-Dimensional Customer Analysis

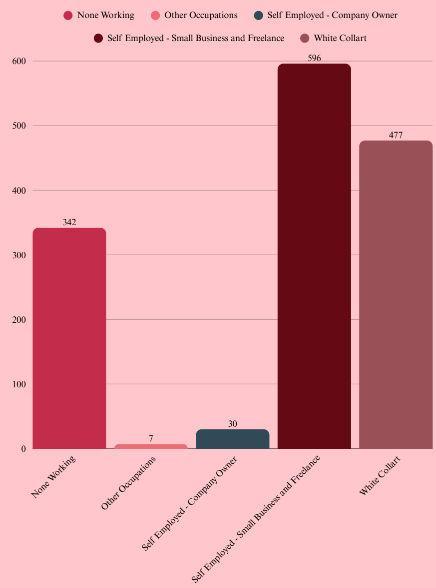
This section presents a comprehensive analysis of customer churn across multiple dimensions, including demographics, brand perception, consumption behavior and usage timing. Each dimension is examined to highlight differences between active and churned customers, thereby identifying key factors that contribute to churn. Additionally, drill-down techniques are employed to explore deeper layers within the data, enabling the identification of high-risk customer segments and their associated characteristics

Demographic Segments Analysis

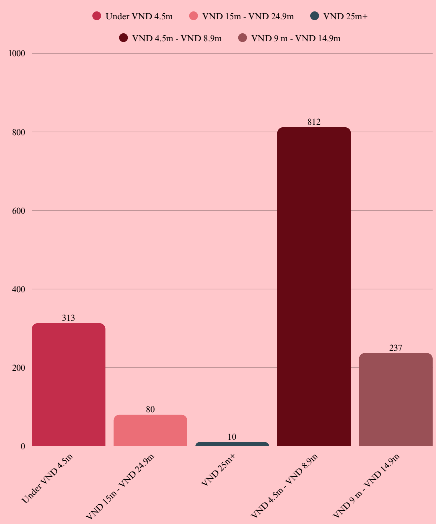
Gender



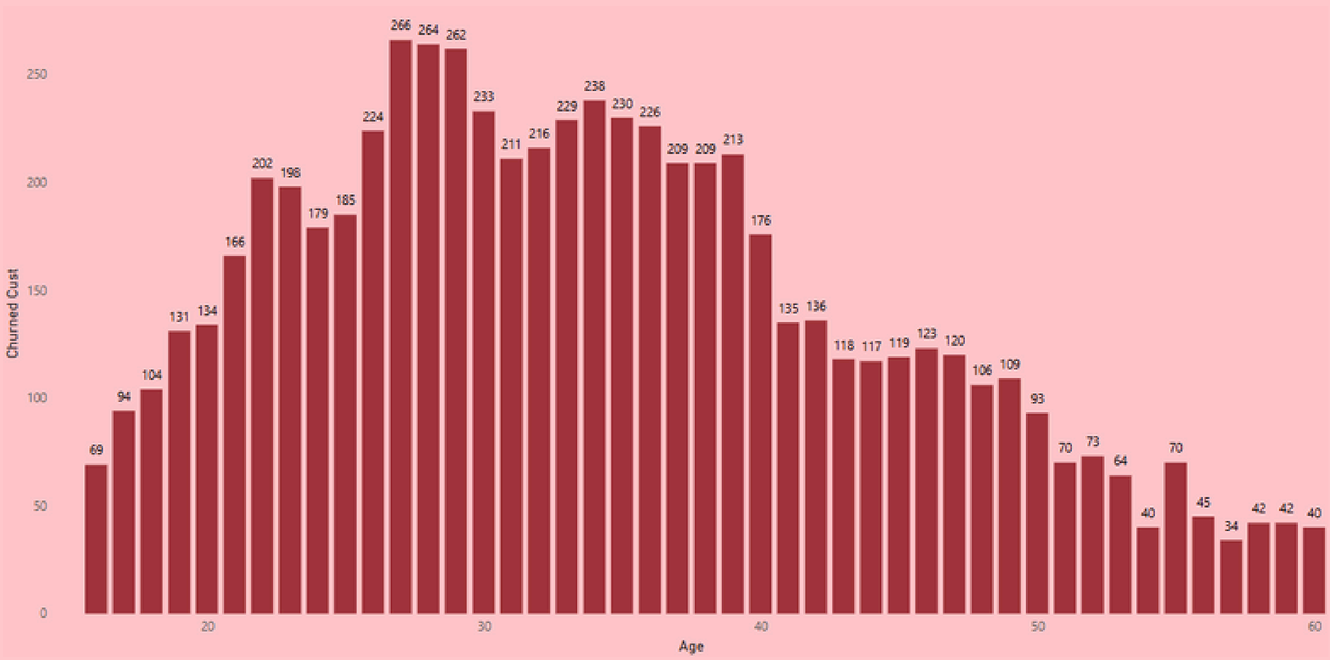
Occupation Group



MPI



Age

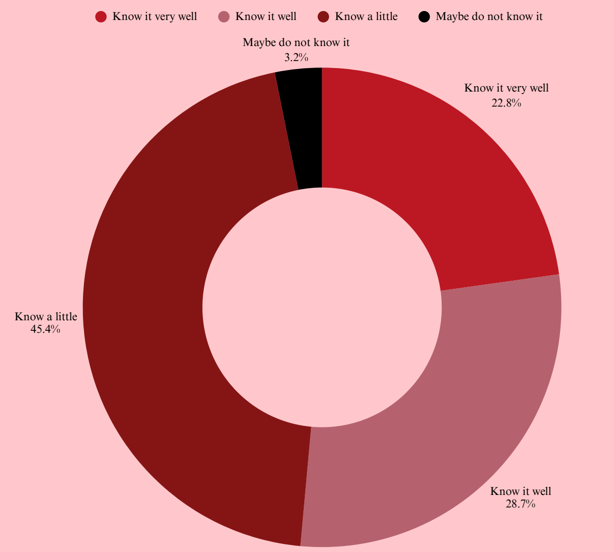


Insights

The demographic profile of churned customers reveals several key patterns. Although the churn rate is slightly higher among females (56.54%), the difference is not substantial enough to suggest gender as a key driver. Age appears more influential, with churn peaking among those aged **28 to 30** — an active consumer segment with high service expectations. This indicates that even core customers may disengage when experiences fall short. In terms of occupation, office staff represent the largest share of churned customers, possibly reflecting a mismatch between expectations and actual service delivery. Churn is also more common among those earning **VND 4.5 to 9 million monthly**, suggesting price sensitivity or perceived value issues. Overall, churn is not limited to peripheral groups but is occurring within essential customer segments, highlighting the need for more targeted retention strategies.

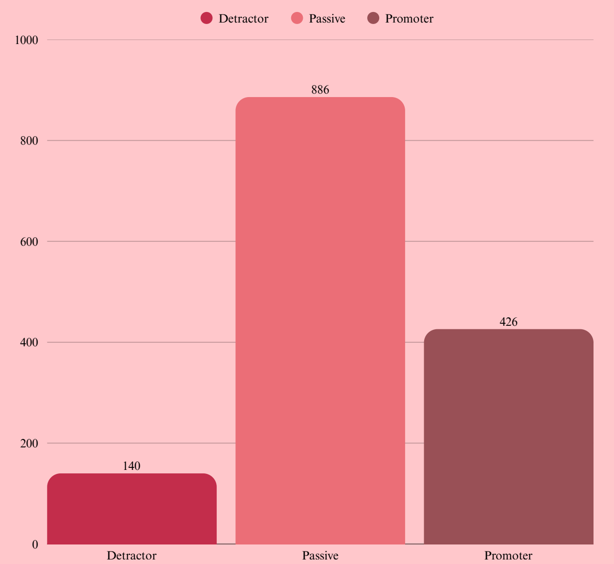
Brand Perception Analysis

Comprehension



The analysis shows that **nearly half** of churned customers have limited understanding of the brand (45.4%), while only **22.8% report a strong level of familiarity**. This lack of awareness may prevent customers from fully recognizing the brand's core value propositions or points of differentiation. The findings suggest an urgent need to strengthen brand positioning and communication, particularly among low-awareness segments, in order to foster stronger brand connection and reduce churn risk.

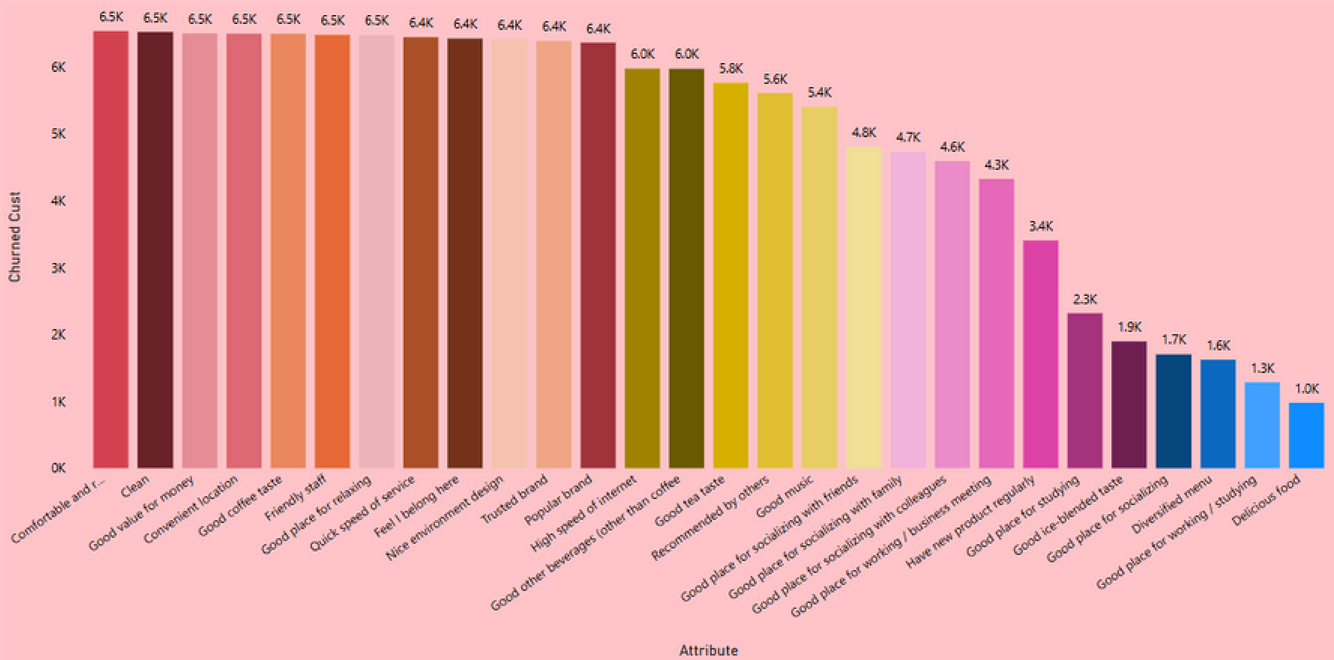
NPS#P3M#Group



Customers classified as **"Passive"** represent the largest share among those who churned (886), significantly outnumbering both "Promoters" (426) and "Detractors" (140). Although passive customers do not express dissatisfaction, their emotional detachment from the brand makes them highly vulnerable to switching. This highlights the need for proactive engagement strategies aimed at converting passive users into loyal advocates, such as targeted communications, reward programs or enhanced customer experiences.

Brand Perception Analysis

Attribute

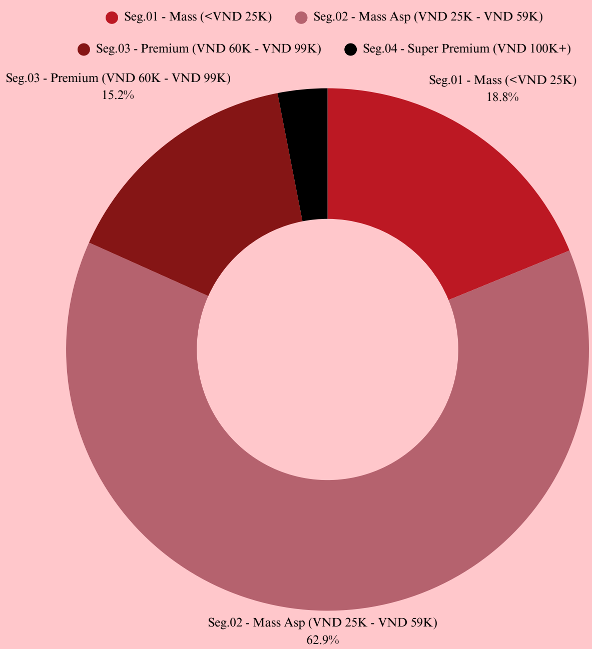


Although churned customers gave high ratings to several individual brand attributes such as cleanliness, value for money, convenience, product taste and service quality, a considerable portion still chose to disengage. Notably, **attributes like "Clean", "Comfortable and relaxing" and "Friendly staff" were each acknowledged by more than 6,500 churned customers**, indicating generally positive perceptions at a feature level.

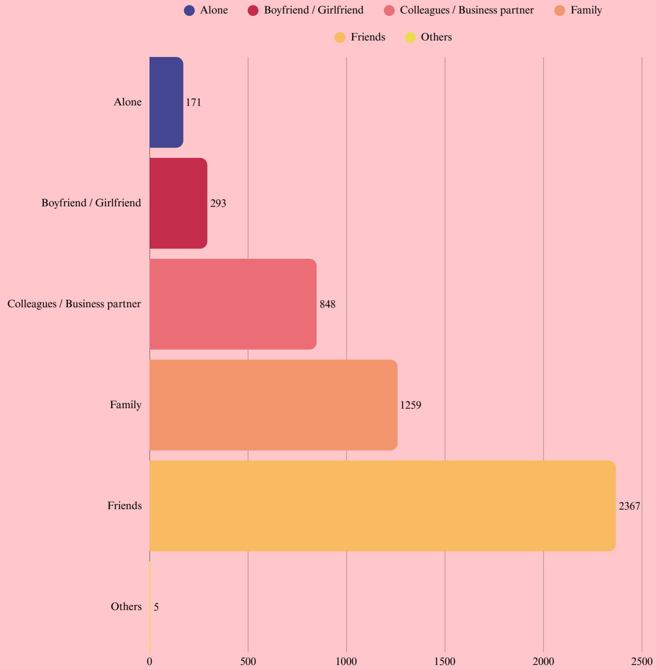
However, this data reveals an important insight: **satisfaction with specific attributes does not necessarily translate into overall loyalty or continued engagement**. Customers may recognize and appreciate individual aspects of the brand experience, yet still decide to leave when the service fails to deliver consistency, emotional resonance or a coherent experience across all touchpoints. This underscores the critical importance of designing a seamless and emotionally engaging customer journey, where individual strengths are integrated into a unified brand experience that fosters sustained loyalty.

Behaviour and Need-State Profiling

Segmentation



Companion#Group

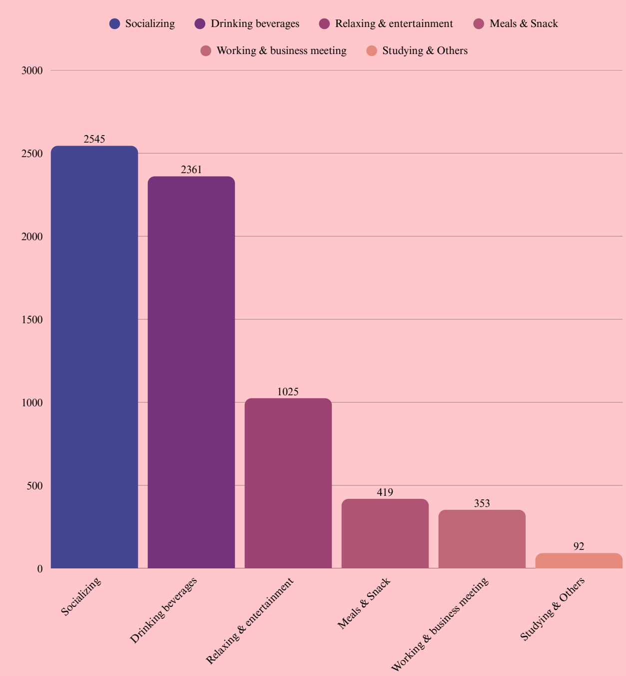


Segmentation data reveals that the **majority of churned customers** fall within **Segment 02 – Mass Aspiration (VND 25K–59K),** comprising 62.9% of the total churn. This mid-tier pricing segment typically includes price-sensitive customers who balance value with experience. The overrepresentation of this group may suggest that current offerings do not fully align with their expectations or that competitors provide stronger value propositions. A deeper understanding of the needs and motivations of this segment is essential to refine positioning, tailor communication, and differentiate service offerings.

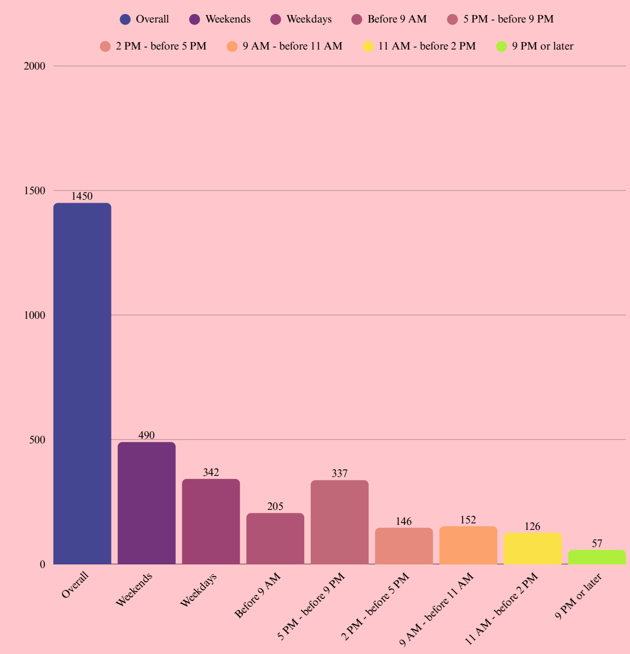
The analysis of companion behavior reveals that **most churned customers tended to visit with others, particularly friends,** who account for 2,367 visits. This is followed by visits with family members at 1,259 and colleagues at 848. In contrast, only 171 customers visited alone. This pattern highlights the influence of social context in shaping café usage, where group-based visits are predominant. Within these settings, even a single negative experience related to service, ambiance, or perceived value can influence the group's collective satisfaction and ultimately contribute to churn. Improving the quality of shared experiences through group-friendly offerings and interactive elements can be an effective approach to strengthen retention among socially driven customers.

Behaviour and Need-State Profiling

NeedstateGroup



Daypart & Dayofweek



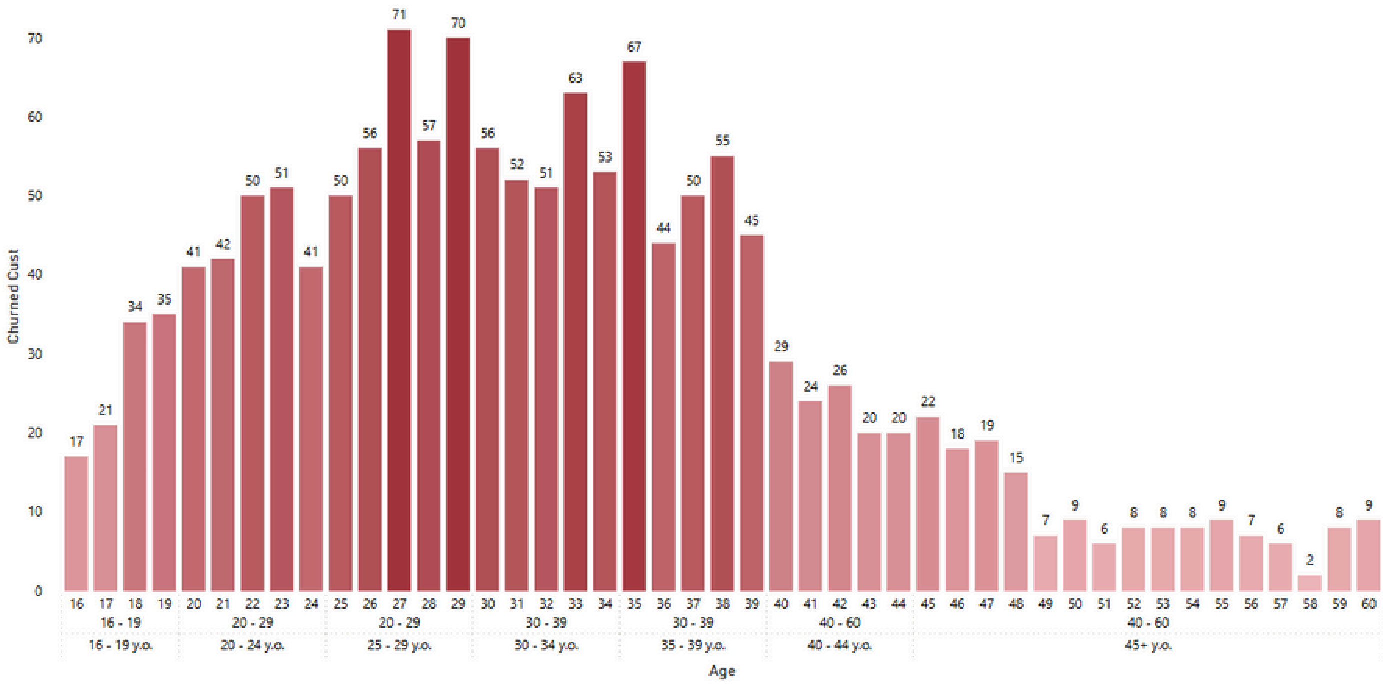
The analysis of customer need states reveals that churn predominantly occurs among individuals engaging in routine and socially driven activities. The largest proportions of churned customers are associated with purposes such as socializing, drinking beverages and relaxing or seeking entertainment. These core usage contexts, which typically represent the brand's primary value propositions, appear to be where expectations are not consistently met. The relatively lower churn among those visiting for work-related meetings or study suggests that specialized experiences may offer more stability. This insight highlights the need for Highlands Coffee to strengthen service delivery and emotional connection in everyday customer journeys, particularly during informal, group-oriented visits where the brand is most exposed.

Temporal patterns in customer churn indicate that the highest rates occur during weekends and in the afternoon to early evening timeframe, specifically **between 2 PM and 7 PM.** The weekend period alone accounts for 490 churned customers, while weekdays see a significant decline in churn rates. These peak windows coincide with periods of intense social activity, where customer expectations regarding service speed, environmental comfort and group facilitation tend to rise. If the brand fails to meet these elevated expectations, especially during such high-demand moments, dissatisfaction may escalate and result in disengagement. It is therefore crucial to prioritize operational efficiency and elevate the customer experience during these critical timeframes in order to improve customer retention.

Hierarchical Drill-down Highlights

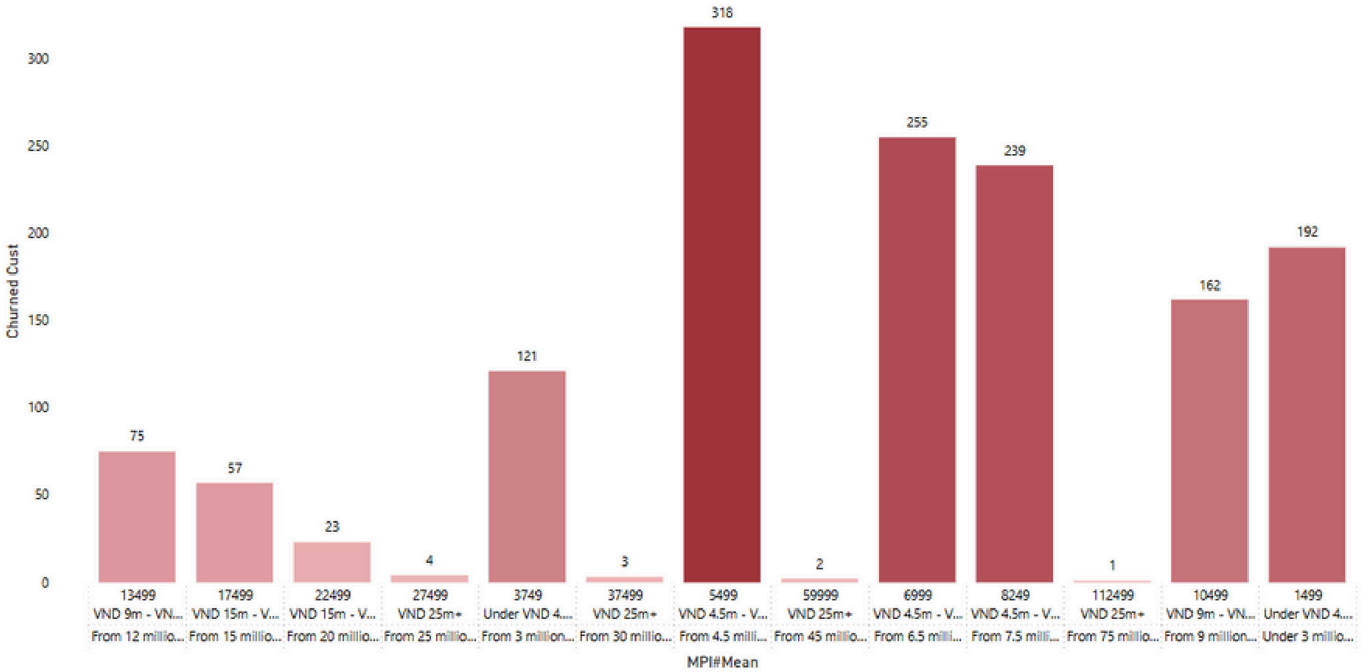
This section investigates the underlying drivers of customer churn through hierarchical drill-down analysis, integrating multi-layered data elements such as granular age bands within broader cohorts, satisfaction scores across NPS segments and usage needs across purpose categories. By examining these nested structures, the analysis reveals latent behavioural patterns that contribute to attrition. The primary objective is to identify significant intersections among demographic, perceptual, and behavioural attributes, thereby establishing a rigorous foundation for more precise and personalised retention strategies.

Age#Group#2 → Age#group → Age



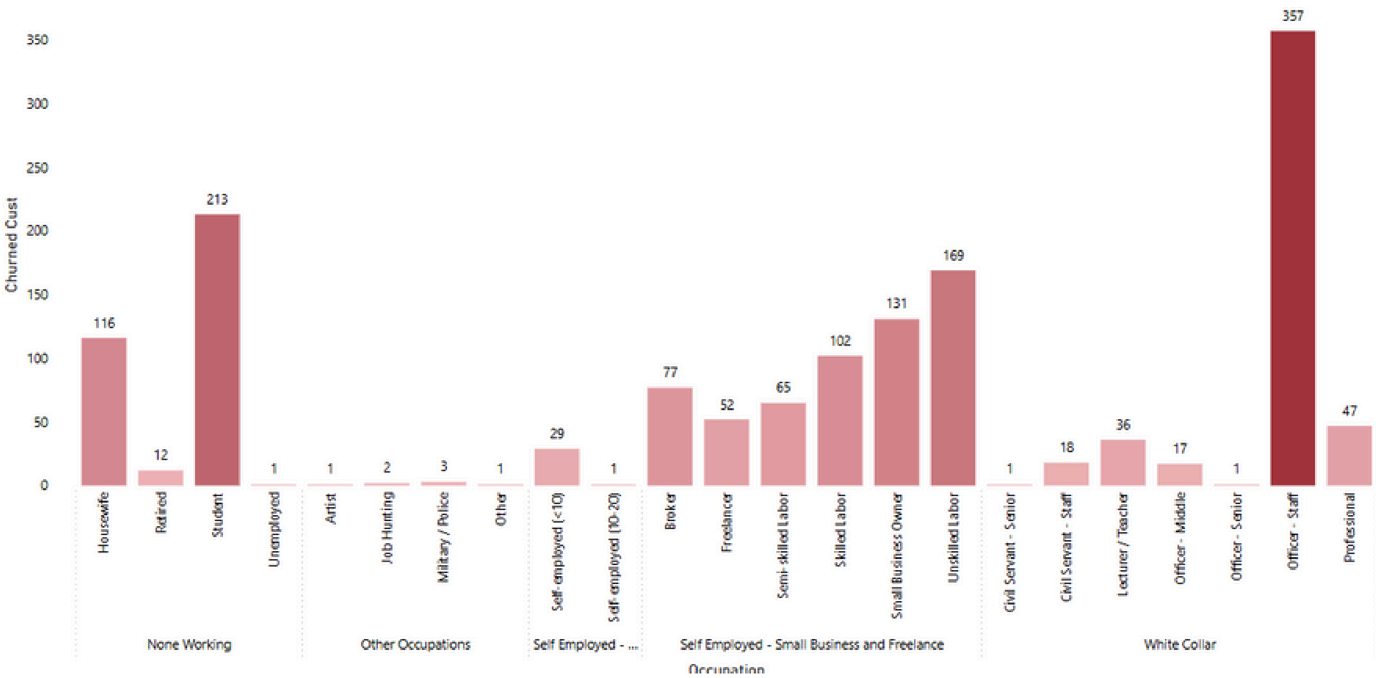
The age-based analysis reveals a notable concentration of churn among customers aged **26 to 38**, with a **pronounced peak between 28 and 30 years old** where churned counts exceed 70 per age. This group represents the core working-age demographic, often characterized by stable income and heightened service expectations. The elevated churn within this segment suggests that current experiences may not be fully meeting their evolving needs. In contrast, customers above 40 exhibit significantly lower churn rates, reflecting more stable behavioral patterns; however, their overall size within the customer base is comparatively smaller.

MPI#detail → MPI → MPI#Mean



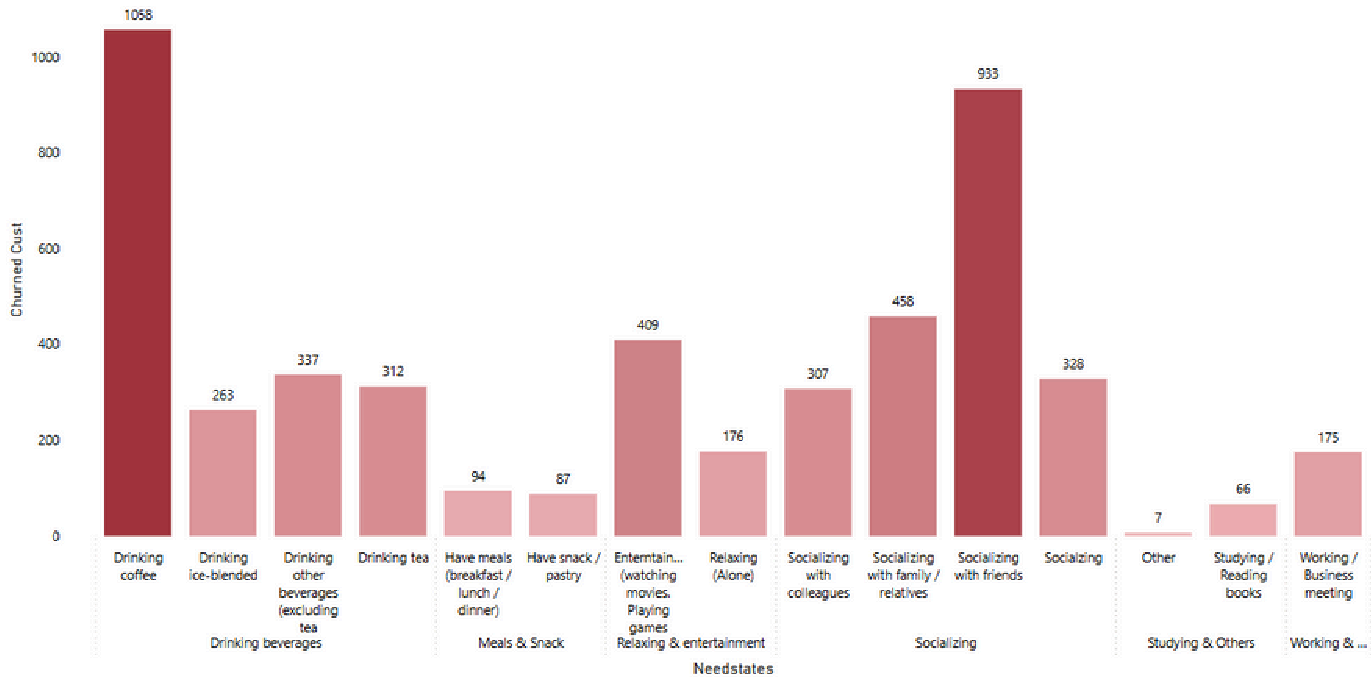
Churn is most common among customers earning around VND 5.5 million per month. These mid-income consumers are more sensitive to value and compare cost with benefit. Although churn is lower among higher-income groups, this may be due to reduced price sensitivity rather than stronger satisfaction. To retain this important segment, pricing and value communication should be improved.

Occupation#group → Occupation



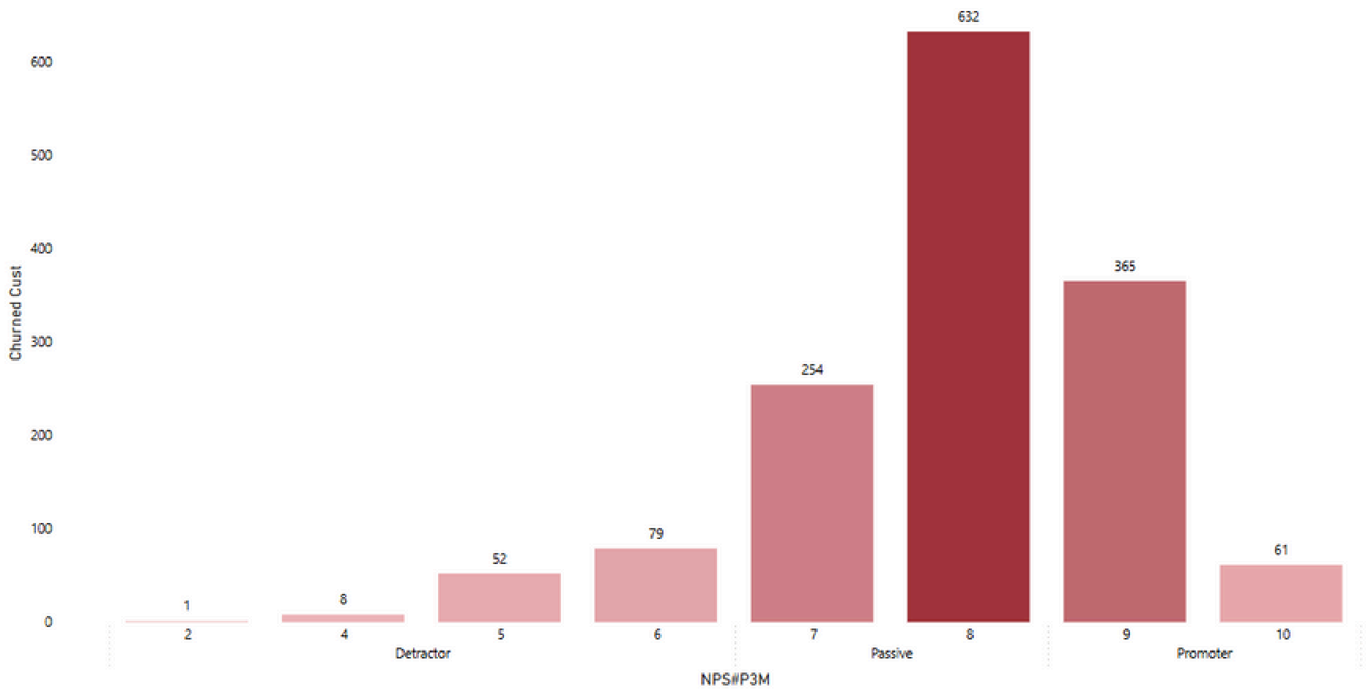
“Office staff” show the highest churn, exceeding all other occupation groups. As this segment tends to expect quality and consistency, the result suggests a gap between service delivery and professional expectations, calling for better-aligned experiences.

NeedstateGroup → Needstates



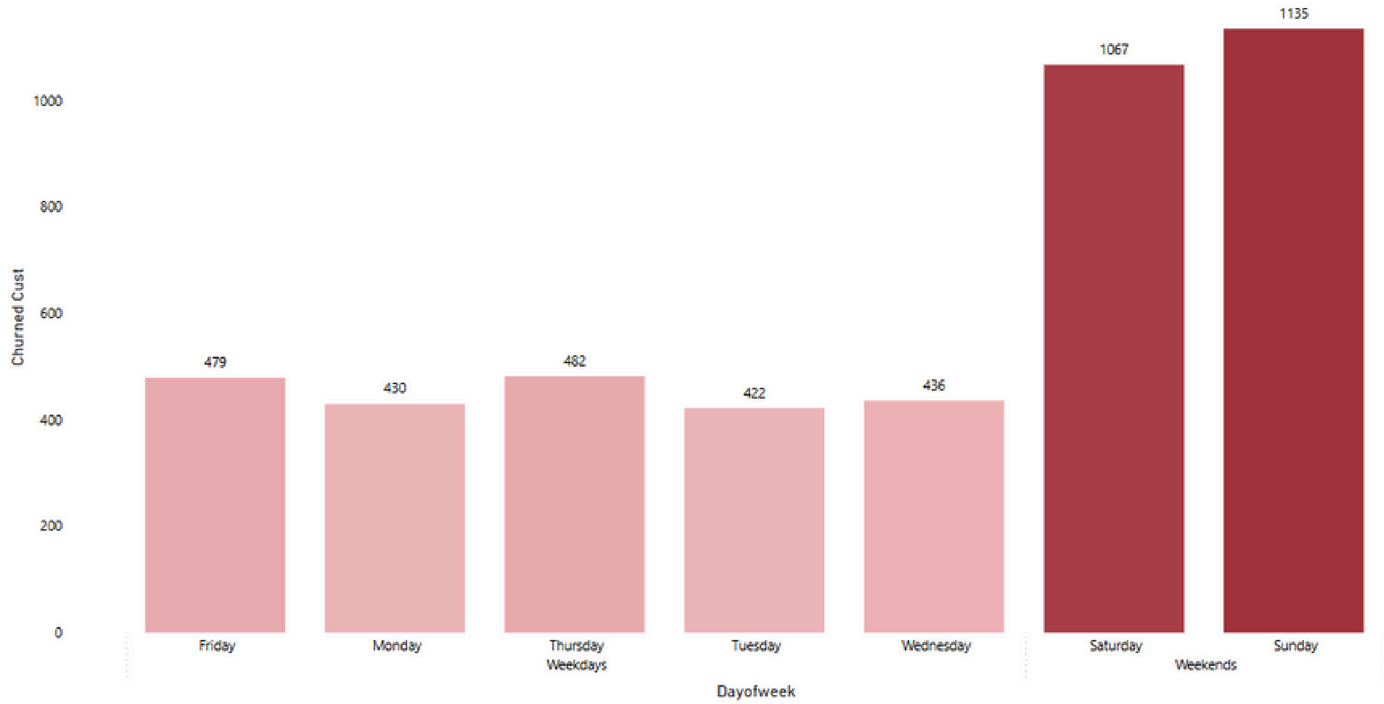
The highest churn occurs among customers who visit for coffee or social gatherings. These are key brand occasions. If the experience in these moments is inconsistent, customers may leave. Improving product consistency and social ambiance is essential to keep these groups loyal.

NPS#P3M#Group → NPS#P3M



Churn is concentrated in the passive group, especially among those who rated the brand an eight. These customers show little emotional commitment and are more likely to leave for better options. Strengthening engagement and building stronger emotional bonds may help convert them into loyal supporters.

Weekday#end → Dayofweek



Churn is highest on weekends, with Sunday and Saturday accounting for the most cases. These are peak periods when customers expect fast service, comfort, and a good group experience. Inconsistencies during these times likely contribute to dissatisfaction. Prioritizing operations and customer care on weekends is crucial for retention.

Conclusion

A comprehensive analysis indicates that customer churn at Highlands Coffee is not confined to peripheral groups but is occurring within the brand's core segments. Notably, young customers aged 26 to 38, those with moderate monthly incomes ranging from 4.5 to 6.5 million VND, and individuals employed as office staff represent the highest proportion of churn. Attrition is also prevalent among customers whose consumption needs align with the brand's central positioning, particularly those who visit for coffee and social interactions. Additionally, **neutral respondents in the Net Promoter Score and customers visiting during weekends appear to be the most vulnerable points in the customer journey**.

These findings suggest that churn is not solely the result of explicit dissatisfaction but may also stem from a lack of emotional connection, inconsistency in the service experience, or unmet expectations. Maintaining a strong brand image is not sufficient on its own unless supported by a consistent and personalized engagement strategy that addresses the full scope of the customer journey.

Recommendation

Enhance experience during peak hour

- Review and optimize service processes during high-traffic periods such as weekends and late afternoons.
- Implement flexible staffing plans and reinforce employee training to ensure consistent service quality and speed.
- Utilize real-time service tracking tools and customer feedback systems to adjust operations dynamically.

Personalize the customer journey and Deepen emotional connection

- Develop a loyalty program tailored to customer behavior and preferences, offering personalized benefits and experiences.
- Use emotional storytelling and brand narrative to reinforce brand identity and build lasting emotional ties.
- Organize community-building activities such as mini-events, group check-ins, or interactive in-store experiences to encourage engagement.

Reposition value for the Middle-income segment

- Design flexible combo offerings that align with spending capacity and convenience-oriented behaviors.
- Strengthen communication around "affordable pricing with high perceived value" to reinforce accessibility.
- Introduce time-based promotions (e.g., lunch-hour deals or low-traffic hour discounts) to maximize value perception.

Reactivate High-risk churn segments

- Develop a loyalty program tailored to customer behavior and preferences, offering personalized benefits and experiences.
- Use emotional storytelling and brand narrative to reinforce brand identity and build lasting emotional ties.
- Organize community-building activities such as mini-events, group check-ins, or interactive in-store experiences to encourage engagement.

Improve brand awareness and understanding

- Expand digital brand presence on high-reach platforms such as Instagram, TikTok and YouTube Shorts.
- Collaborate with influencers and media partners to increase brand visibility and trust.
- Develop educational content that effectively communicates the brand's core values, quality standards, and story in an engaging and accessible format.

PART 06

CUSTOMER SEGMENTATION AND CHURN PREDICTION

Customer SegmentationModel

Feature selection and Data preparation

Variable	Datatype
Fre#visit	float64
Spending	float64
NPS#P3M	float64
Companion#group	object → Label Encoding
Daypart	object → Label Encoding
NeedstateGroup	object → Label Encoding

Data Scaling using StandardScaler()

For the customer segmentation model, we use the following columns to segment based on customer ***behavior, needs, and satisfaction:***

- **Fre#visit** reflects the frequency of customer visits, helping distinguish "loyal" customers from "trial" customers, and guiding retention and re-engagement strategies.
- **Spending** allowing us to separate "high-value" customers from "low-value" customer and, when combined with Fre#visit, to identify groups like "frequent but low spend" or "infrequent but high spend."
- **NPS#P3M** measures satisfaction and likelihood to recommend, serving as a key psychological indicator to distinguish "promoters" (high NPS) from "detractors" (low NPS) for tailored customer care.

- **Companion#group** shows who the customer came with (alone, friends, family, etc.), providing insight into social context and shopping behavior (for example, customers accompanied by family often spend more).
- **Daypart** indicates the time of day for each visit (morning, afternoon, evening, night), helping us understand time-based habits and optimize campaigns for peak and off-peak hours.
- **NeedstateGroup** represents the primary need or motivation for each visit (information gathering, entertainment, urgent purchase, etc.), enabling segmentation by visit purpose and personalization of messaging and product recommendations.

Building Clustering Model and Result

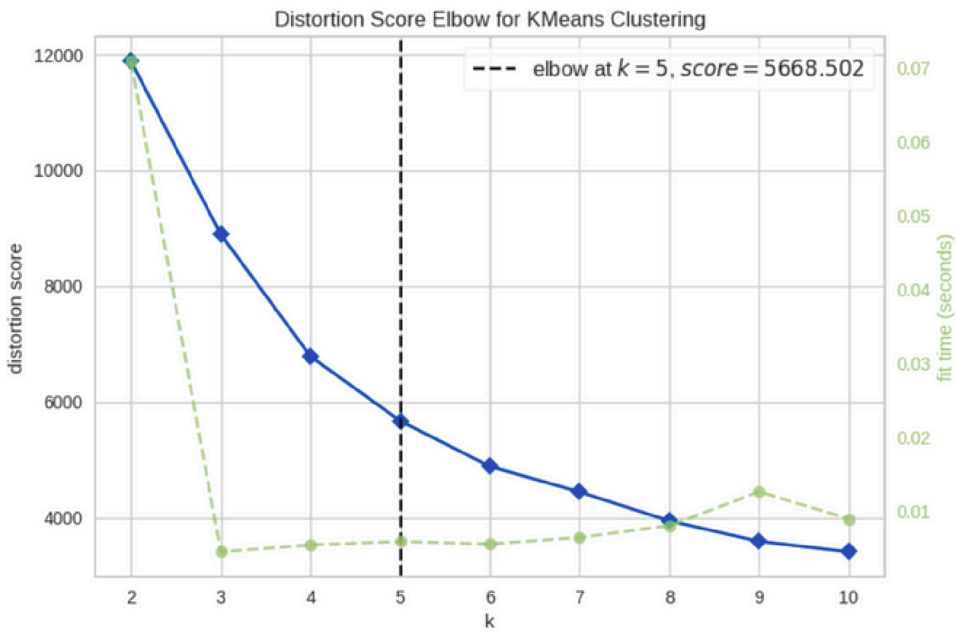


Figure: Elbow Method to determine the number of clusters

We used the **Elbow method** on the K-Means distortion plot to pinpoint the optimal cluster count, noticing a clear "knee" at **k = 5** where the distortion score levels off. This data-driven inflection confirms that five clusters capture most of the structural variance without overfitting.

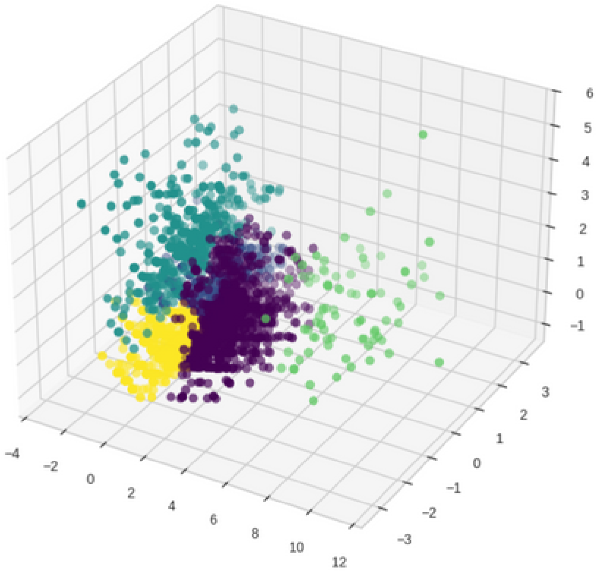
	count	mean	std	min	25%	50%	75%	max
col1	4146.0	0.000000e+00	1.381536	-3.244468	-0.850573	-0.374114	0.491626	11.357336
col2	4146.0	0.000000e+00	1.031910	-3.266607	-0.748680	0.058353	0.687432	3.461847
col3	4146.0	4.113127e-17	1.005657	-1.067142	-0.551966	-0.363489	-0.048958	5.539450

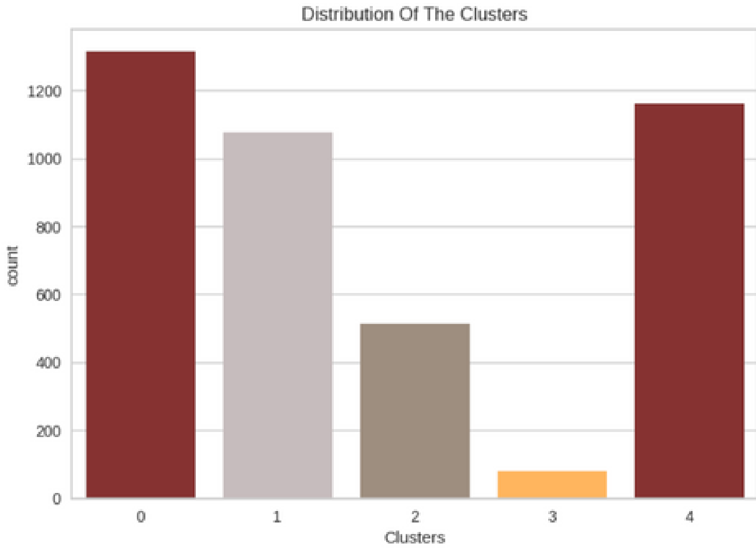
Figure: Descriptive Statistics for PCA-Transformed Features

After reducing our six key features into three principal components, the customer points in PCA_ds form irregular, cloud-like groupings with long tails and outliers (PC1 spans from 3.24 to 11.36). K-Means forces spherical clusters that cut through these shapes, and DBSCAN either merges sparse segments or treats them as noise when tuning density parameters. **Agglomerative Clustering**, however, iteratively merges the closest points based purely on distance, so it naturally **respects the true contours and varying densities of our data without imposing uniform cluster shapes.**

Moreover, the dendrogram generated by **Agglomerative Clustering shows a pronounced jump in linkage distance** just before the fifth cluster is formed, giving us a clear, data-driven rationale to set n_clusters=5. This hierarchical view **makes the segmentation more interpretable**—each of the five clusters (e.g "high-value promoters," "late-night bargain hunters") can be justified by its merge distance - helping stakeholders understand and trust the resulting customer groups.

Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
1313	1078	512	81	1162
31.67%	26%	12.35%	1.95%	28.03%





Clusters	Fre#visit	Spending	NPS# P3M	Companion #group	Daypart	Needstate Group
0	5.44	215.32	9.27	Family	5 PM - before 9 PM	Drinking beverages
1	1.39	21.91	7.88	Friends	5 PM - before 9 PM	Drinking beverages
2	2.36	43.68	8.09	Friends	5 PM - before 9 PM	Socializing
3	19.78	952.86	9.33	Colleagues / Business partner	5 PM - before 9 PM	Drinking beverages
4	1.64	32.86	8.21	Colleagues / Business partner	11 AM - before 2 PM	Drinking beverages

Cluster 0 – Stable Family Visitors: This segment accounts for the largest portion **(31.67%)** of all customers. They have a high average visit frequency (**5.44 times per 3 months**) and a moderate average **spending of 215.32 per visit**. Their average Net Promoter Score **(NPS) is also high (9.27)**, indicating strong satisfaction and likelihood to recommend. The most common visit context involves **family members**, primarily during the **5 PM – 9 PM time** window, and the main needstate is **drinking beverages**. This is a stable and loyal segment with long-term retention potential - well-suited for loyalty programs or family-focused offers.

Cluster 1 – Budget-Conscious Friend Groups: Comprising **26%** of the customer base, this group has the lowest average **visit frequency (1.39 times)** and **spending (21.91 per visit)**. However, their average **NPS remains positive (7.88)**. They typically visit with **friends** in the evening **(5 PM – 9 PM)** for **drinks**. Despite low spending, this sizable segment could be activated through promotions, group discounts, or "bring-a-friend" campaigns that increase both frequency and value.

Cluster 2 – Socializing Friends: Making up **12.35%** of the base, these customers have moderate visit **frequency (2.36)** and slightly higher average **spending (43.68)** compared to Cluster 1. The average **NPS is 8.09**, indicating decent satisfaction. Like Cluster 1, they also come with **friends** during the evening, but their most common needstate is **"socializing"** rather than simply drinking. Offering a more interactive, experience-driven environment may better attract and retain this segment.

Cluster 3 – Premium Business Professionals: Though they represent only **1.95%**, this is the most high-value cluster: they **visit frequently (19.78 times per 3 months)** and spend significantly more than any other group (**952.86** per visit). Their average **NPS is the highest (9.33)**. They usually come with **colleagues or business partners,** also in the evening, and typically for drinks. This elite customer group should be treated as VIPs—with personalized services, exclusive offers, and premium in-store experiences.

Cluster 4 – Daytime Business Visitors: This segment accounts for **28.03%** of customers and has a low average **visit frequency (1.64)** and moderate average **spending (32.86)**. They usually visit during midday hours **(11 AM – 2 PM)** with **business colleagues**, also primarily for drinks. With an average **NPS of 8.21**, they express decent satisfaction. This cluster presents an opportunity to offer streamlined, professional lunch or drink combos tailored to the workday crowd.

Overall, the analysis reveals five distinct customer segments differentiated by behavior, spending, and motivations. Clusters 0 and 3 emerge as the most valuable - either due to their volume (Cluster 0) or high spending and frequency (Cluster 3) - and should be prioritized with retention strategies and premium offerings. Clusters 1 and 2, despite lower spending, are socially engaged groups that can be reactivated through group promotions or interactive experiences. Cluster 4 represents a professional, daytime clientele that can be tapped with efficient service and business-friendly deals. Tailoring strategies based on visit timing, companion type, and primary needstate will allow businesses to personalize engagement and maximize value across segments.

Recomendation

Stable Family Visitors	Budget-Conscious Friend Groups	Socializing Friends	Premium Business Professionals	Daytime Business Visitors
• Weekend Package: Food and Drink Combo for 4–6 people at fixed price, free dessert for children • Special Event: Organizing "Family Time" on Sunday afternoon: mini games for children, painting corner, 20% discount for large families.	• "Bring - a - Friend" Campaign: When a group of 3–4 friends arrive, get 10–15% off the total bill if you invite a new guest. • Sharing Combo: Set of 4 soft drinks at a discounted price, plus a "buy 2 get 1 free" promotion for snacks.	• "Instagrammable" space: Beautiful corner design, unique backdrop for young people who love taking pictures to share. • Club membership: "Social Club" membership card with the right to participate in pre-sale events, 15% discount on any drink.	• VIP Lounge & Private Reservations: Quiet Zone, Air Purifier, Wireless Charging, Private Seat Reservations. • B2B Partnerships: Corporate Partnerships, Free Trial Packages for Partners; VIP Customer Service Program.	• Lunch Express Combo: Lunch set + drinks from 11 AM–2 PM, delivered within 10 minutes • Company Membership: Card for nearby office workers, get a fixed 10% discount each time

Limitation

In this report, the customer segmentation model is built on three core metric groups—behavior (visit frequency, spending), needs (beverage preferences, socializing) and satisfaction (NPS)—which effectively identify clusters with similar characteristics. However, the model does not fully capture qualitative and demographic dimensions such as age, income and geographic location; it also lacks data on interaction channels (online vs. offline), detailed product preferences and seasonal factors. Incorporating these additional data points would enhance the model's accuracy, flexibility and responsiveness to market shifts, thereby optimizing targeting and service strategies for each customer segment.

CHURN PREDICTION MODEL

Objective

This modeling exercise aims to predict customer churn risk at Highlands Coffee based on behavioral and demographic features. The goal is to enable early identification of high-risk individuals and support personalized retention strategies, thereby improving customer lifetime value and business sustainability. This initiative not only enhances customer engagement efforts but also contributes to the long-term financial resilience of the brand by proactively addressing potential customer attrition.

The binary target variable churn_flag is constructed based on customer activity:

- **churn_flag = 1** if a customer has made purchases in the past 3 months (P3M) but has not visited in the last month (P1M), indicating disengagement.
- **churn_flag = 0** if the customer has made at least one visit in the last month (P1M).

This definition aligns closely with business logic and operational relevance. It ensures that the model targets meaningful churn behavior rather than incidental inactivity, providing actionable insights for marketing and operational interventions.

Feature selection and Data preparation

Six predictor variables were selected based on relevance to customer behavior and segmentation:

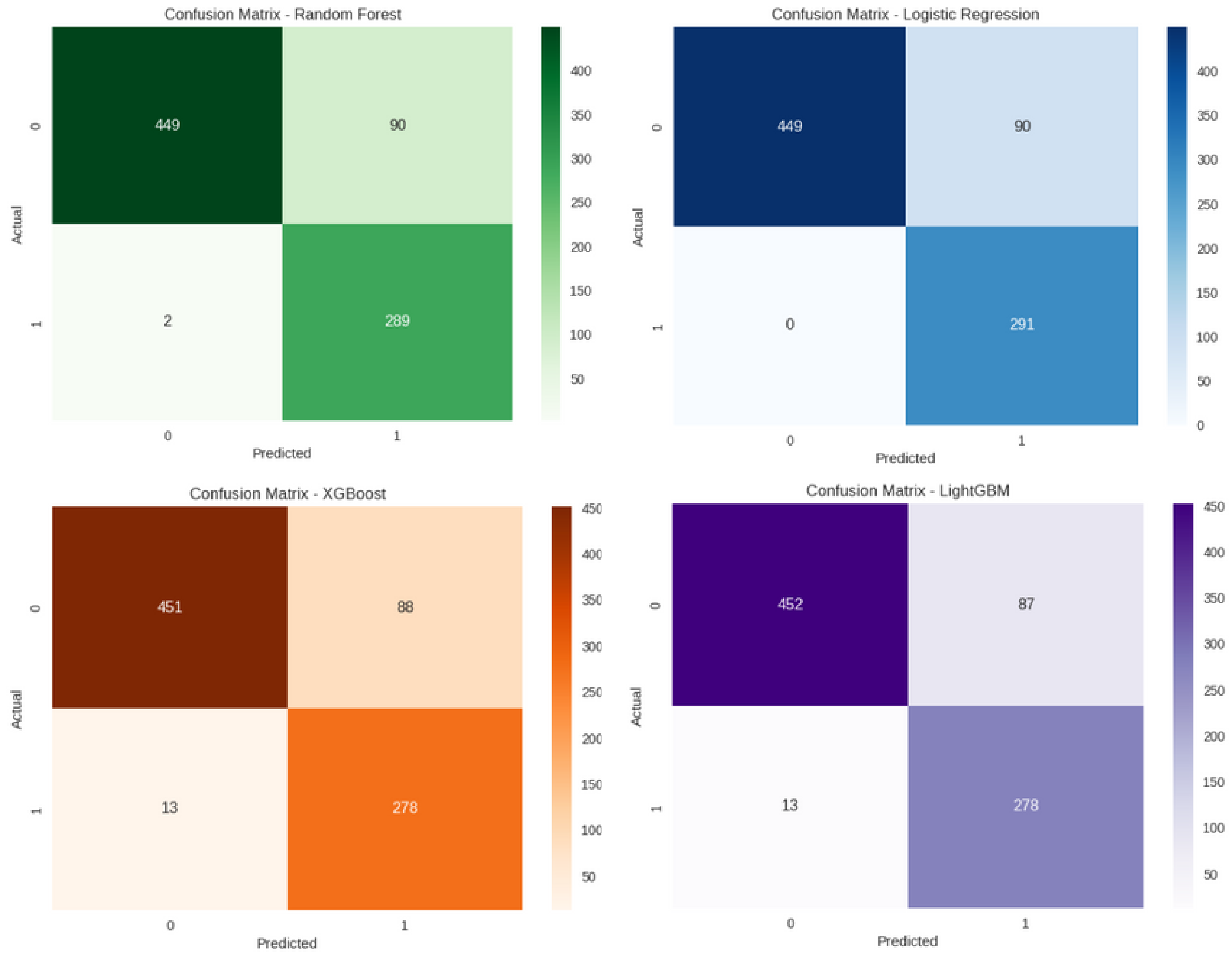
- Frequency of visit (Fre#visit)
- Need state group (NeedstateGroup)
- Companion group (Companion#group)
- Time of day (Daypart)
- NPS (NPS#P3M)
- Predicted cluster (Clusters)

Categorical features were transformed **using label encoding** to ensure compatibility with machine learning algorithms. Missing values were handled by removing incomplete records to maintain data integrity. After cleaning, the final dataset contained **830 complete samples**, offering sufficient balance across classes for reliable training. **Data was split into training and testing subsets using an 80-20** stratified approach to preserve class distribution and ensure representative evaluation performance.

Model training and Evaluation

Four classification algorithms were employed to model churn risk: Random Forest, Logistic Regression, XGBoost and LightGBM. Each model was evaluated on accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix to comprehensively assess predictive performance.

Confusion matrix visuals were generated to validate prediction stability across models.



The table below summarizes the key evaluation results:

Model	Accuracy	ROC-AUC	Precision (1)	Recall (1)	F1-Score (1)
Random Forest	0.89	0.9298	0.76	0.99	0.86
Logistic Regression	0.89	0.9129	0.76	1.00	0.87
XGBoost	0.88	0.9258	0.76	0.96	0.85
LightGBM	0.88	0.9275	0.76	0.96	0.85

This implies that

Although all four models performed well, Logistic Regression was selected for deployment due to its strong balance of recall, transparency and business applicability. While Random Forest achieved the highest ROC-AUC (0.9298), Logistic Regression offered perfect recall (1.00) for churners, which is critical for minimizing false negatives and enabling timely retention efforts.

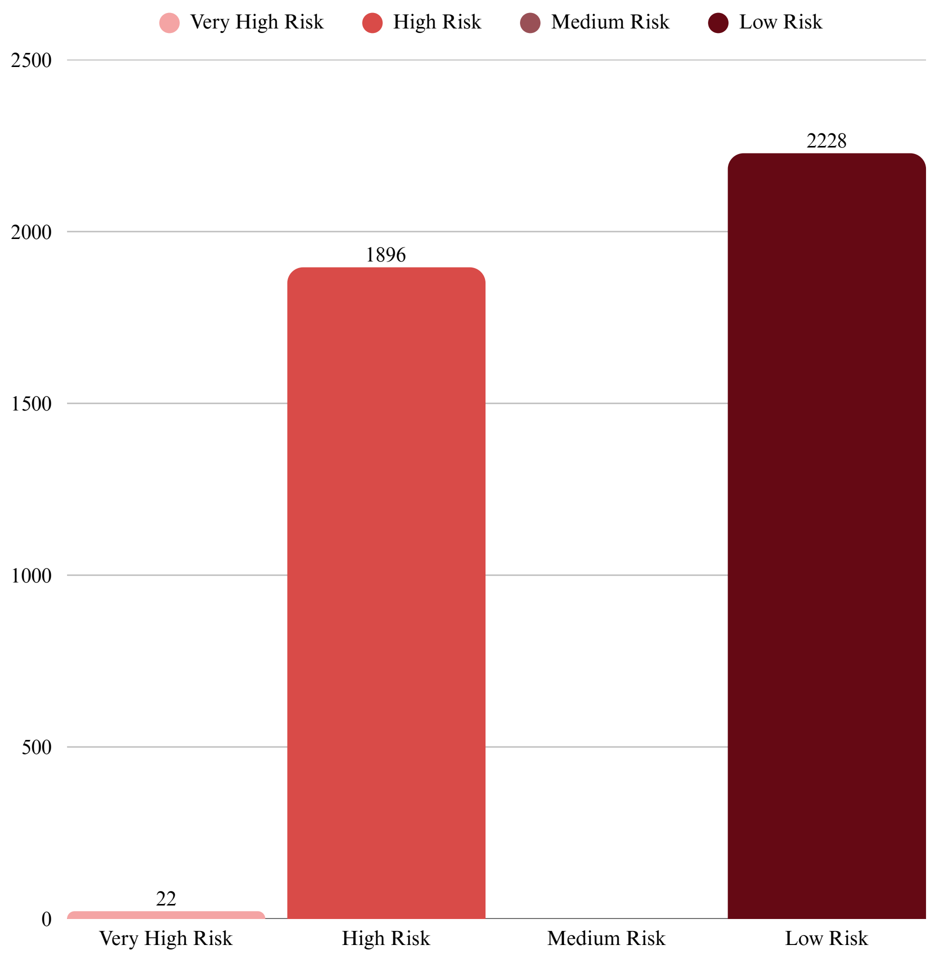
Its interpretability also makes it easier to communicate insights to business stakeholders and integrate into operational workflows. While ensemble models like XGBoost and LightGBM had similar accuracy, their complexity limits transparency and real-time use.

Predict churn risk across all customers

The final model was applied to the entire customer database to estimate individual churn probabilities (churn_proba). Customers were categorized into four risk levels based on these probabilities:

- Very High Risk: 0.9 and above
- High Risk : 0.6 to under 0.9
- Medium Risk : 0.3 to under 0.6
- Low Risk : below 0.3

A bar chart was created to illustrate the distribution of customers across these tiers. This visualization helps identify where proactive engagement efforts should be prioritized.



The majority of customers fall into the Low Risk tier, indicating a strong engagement base. However, the large proportion in the High Risk group signals a significant opportunity for retention intervention. Though small in size, the Very High Risk segment requires urgent attention due to imminent churn likelihood. This distribution highlights the need to prioritize outreach to high-risk tiers while maintaining satisfaction among loyal customers.

Feature importance

The deployed Logistic Regression model provides fully interpretable coefficients that quantify how each predictor influences the log-odds of churn.

Feature	Coefficient	Interpretation
Fre#visit	-5.854	More visits strongly reduce churn risk
Clusters	0.219	Certain behavior-based clusters increase risk
Daypart	0.154	Specific times of day correlate with higher churn likelihood
NPS#P3M	0.044	Neutral sentiment modestly raises risk
Companion#group	0.024	Social context has limited but positive influence on churn
NeedstateGroup	0.017	Underlying consumption need has minimal effect

➜ **Insight 1** - Habitual engagement is the single most powerful safeguard against churn. Encouraging even one additional monthly visit can materially lower risk.

➜ **Insight 2** - Temporal and clustering variables contribute meaningfully but are an order of magnitude less influential than frequency, suggesting they should refine, not drive, retention tactics.

Conclusion

The churn prediction model achieved high performance across all tested algorithms. Logistic Regression was selected as the optimal solution due to its high recall, interpretability, and deployment readiness. The findings suggest that both behavioral timing and social engagement are key dimensions to address for effective churn management.

Recommedation

The deployed Logistic Regression model provides fully interpretable coefficients that quantify how each predictor influences the log-odds of churn.

01

Target high-risk segments for retention

- Focus on customers categorized as "Very High" and "High Risk".
- Leverage churn probabilities to personalize incentives and increase return visits.

02

Optimize operations during critical periods

- Enhance service consistency during peak times such as weekends and afternoon hours.
- Ensure operational readiness for high-traffic time slots.

03

Reinforce the social experience

- Develop promotions tailored to group visits based on companion behavior.
- Adjust store layout and ambiance to foster longer, shared experiences.

04

Maintain model performance and integration

- Regularly retrain the model using updated behavioral data.
- Integrate churn predictions into CRM systems for real-time intervention and monitoring.

Limitation

Despite the model's strong performance, several limitations suggest opportunities for refinement. First, the dataset omits key behavioral variables such as transaction value, dwell time, or digital engagement, which could enhance predictive accuracy. Second, the fixed churn definition may overlook individual lifecycle differences, and assuming constant feature effects may oversimplify behavior. Third, external influences like promotions or economic conditions were not included, potentially limiting the model’s generalizability. Future efforts should enrich the dataset with granular and temporal features, explore more flexible churn definitions (e.g., survival analysis), and integrate contextual variables to strengthen robustness and actionable precision.

GÀ MỜ CÓ SỐ

Email

nhuquynh211104@gmail.com
Team Leader

Address

University of Economics and Laws